Cockroach Farming Business Plan: East Asia, 39–40°N

Nov 8, 2026 · @Flyn Rider

Executive summary

We ask for CNY 100,000 (USD 15,000) for a small dubia pilot farm. It will be in northern Hebei or outer Tianjin. After that, we ask for CNY 2.5 million (USD 375,000) for a medium farm. We build the medium farm only if the pilot passes three written tests (gates). At our careful output of 1.2 kg of live insects per m² a month, the pilot earns back its money in about 24 months. The medium farm earns back its money in about 54 months, but only at 2.0 kg/m². At 1.2 kg/m² it never earns it back. This is the main finding of the plan. Medium and large farms are a good investment only after a small room proves the output (yield) and the buyers.

This plan covers the 39–40°N latitude band of East Asia. This means northern China (Beijing, Tianjin, northern Hebei, southern Liaoning, northern Shanxi and southern Inner Mongolia), North Korea (Pyongyang to Hamhung) and northern Japan (Iwate and Akita). South Korea is just south of 39°N. The whole band has the same farming problem. January is cold, with an average of −6 to 0 °C. So a farm for tropical cockroaches must heat for five to seven months a year. Summers reach 35–42 °C, so the farm must also cool.

Investment ask

Item	Phase 1: small pilot	Phase 2: medium farm (only after the tests are passed)
Amount sought	CNY 100,000 (USD 15,000)	CNY 2.5 million (USD 375,000), at least half as owner money (equity)
Use of funds	Equipment and room set-up CNY 70,840. Early losses until cash is at its lowest point about CNY 7,900. Reserve about CNY 21,300	Set-up and equipment CNY 1,724,000. Rent deposit CNY 31,200. Running losses until cash is at its lowest point about CNY 666,000 (at 2.0 kg/m²). Reserve about CNY 79,000
Site	60 m² insulated room in an existing building	1,000 m² of rearing rooms in a leased 1,300 m² rural building

Item	Phase 1: small pilot	Phase 2: medium farm (only after the tests are passed)
Milestones	Month 4 first sales. Month 15 full output. Month 24 all money earned back at 1.2 kg/m²	Month 5 first sales. Month 26 full output. Month 54 all money earned back at 2.0 kg/m²
Test (gate) to start the next phase	Sell 95% of output for six months. Produce at least 1.9 kg live/m² a month. Sign buyers for two-thirds of medium farm output at an average price of CNY 110/kg or more	Two years of profit at medium size, and contracts for most extra output, before any large farm
Return, careful case (1.2 kg/m²)	Operating profit (EBITDA) CNY 67,000 a year, owner not paid. Money earned back in month 24. Total cash +CNY 203,000 by month 60	Operating profit CNY 35,000 a year. Money never earned back. Total cash −CNY 2.65 million by month 60
Return, good case (2.0 kg/m²)	Operating profit CNY 124,000 a year. Money earned back in month 17. +CNY 448,000 by month 60	Operating profit CNY 871,000 a year. Money earned back in month 54. +CNY 468,000 by month 60

All returns are estimates from this plan's model. They are monthly, before tax, before depreciation and before loan costs, at northern Hebei costs. The owner runs the pilot himself. If the owner takes a wage of CNY 5,600 a month, the pilot only breaks even (no profit, no loss) at 1.2 kg/m². At 2.0 kg/m² it earns back its money in month 37. For an investor, the pilot is useful for the proof it gives, not for its profit.

Key findings

- **Best place to start: a live-feeder dubia (杜比亚) farm in northern Hebei or outer Tianjin**. It is 1–2 hours from the Beijing and Tianjin reptile markets. Power at the farm price costs about CNY 0.47–0.63 per kWh, based on the local grid. Hebei's minimum wage is CNY 2,080–2,380 a month. Farm workers cost CNY 3,000–5,500.
- **Small farm (our estimate):** a 60 m² room costs about CNY 70,800 (USD 10,600) to equip. It earns about CNY 67,000–124,000 a year before the owner is paid. It earns back its money in 17–24 months if dubia wholesale prices stay the same. It still makes cash even if too much supply cuts prices by 35%.
- **Medium farm:** about 1,000 m² of rearing rooms. It needs CNY 1.7 million of start-up cost and CNY 2.4 million of money at the highest point. It earns back its money in about 4.5 years at 2 kg of live insects per m² a month. But operating profit (EBITDA) is

close to zero at 1.2 kg/m². It is below zero if prices fall by one-third. Build it only after the small farm proves output and sales.

- **Large farm (4,000 m², CNY 5.9 million of capital):** does not earn back its money within five years in our model, even at 2 kg/m². Build it only with signed contracts for wholesale or freeze-dried sales.
- **Ground beetle cockroach (土鳖虫):** the only cockroach in the official Chinese medicine list (Pharmacopoeia). It comes from Hebei (冀地鳖). It is sold at the Anguo herb market south-west of Beijing. At 2026 prices of CNY 35–45/kg dried, it pays only as small extra work for a family. Heating at 39–40°N makes it worse.
- **Avoid:** American cockroach for medicine makers without a contract (the four Kangfuxin makers farm their own). Kitchen-waste plants (the best-known one, in Zhangqiu, failed). Cockroach powder sold as food (it is not an approved food).
- **Japan (Iwate, Akita):** legal, with good prices for live feed insects. But workers (JPY 1,090/hour minimum) and power cost much more. It suits only a small farm for high-price products. **North Korea:** UN sanctions (trade bans) stop joint companies, machine supply and food exports. We cannot work there.

Key words

Word	Meaning in this plan
EBITDA	Operating profit: profit before tax, interest and depreciation.
Cash flow	Money coming in minus money going out in a period.
Break-even	The point where income equals costs: no profit, no loss.
Time to earn back the money	The time it takes to earn back the money put in.
Equity	Money from owners or investors, not a loan.
Gate	A test that must be passed before the next step starts.
Yield	Output: kg of live insects per m² a month.
Wholesale	Selling large amounts to shops or traders at a lower price.
Retail	Selling small amounts directly to end buyers.

Word	Meaning in this plan
Freeze-dried	Dried by freezing, so the insect keeps a long time.
Pharmacopoeia	The official national list of approved medicines and medicine materials.
GAP	Good Agricultural Practice: official farming quality standard for medicine materials.
VAT	Value-added tax: a tax added to the sale price.
Sanctions	International rules that ban trade and business with a country.
Frass	Insect droppings, which can be sold as fertiliser.
Dubia	Blaptica dubia, a tropical cockroach sold as live food for reptiles (杜比亚).
Nymph	A young insect that is not yet adult.

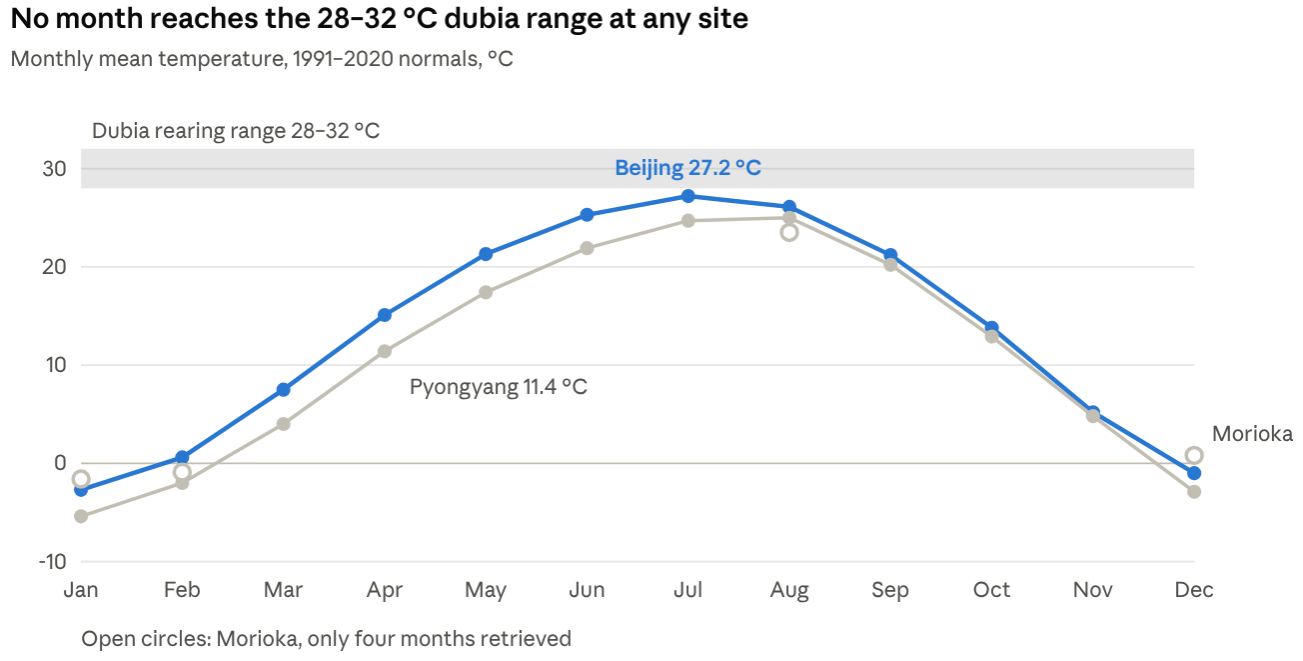
The band: places and climate

Dubia need 28–32 °C. At every possible site, the average temperature of every month is below this. So a farm anywhere in the band must heat for about seven months (October–April). It must also heat on cool nights in May and September. City heating seasons are shorter than the farm's. They are 15 November to 15 March in Beijing and Tianjin, and 5 November to 5 April in Dalian (市井生活58, self-media).

Place	Country	Latitude	January average	Hottest month average	Lowest / highest ever	What it means
Beijing	China	39.9°N	−2.7 °C	27.2 °C (July)	−27.4 / 41.9 °C	Heat October–April. Cool July–August
Tianjin	China	39.1°N	−2.8 °C	27.2 °C (July)	−22.9 / 40.6 °C	Same as Beijing. Land and workers cost less
Langfang	China	39.5°N	−3.4 °C	26.9 °C (July)	−21.8 / 41.4 °C	Between Beijing and Tianjin. Hebei wages
Tangshan	China	39.6°N	−4.6 °C	26.4 °C (July)	−25.2 / 40.1 °C	Colder winters than Beijing
Qinhuangdao	China	39.9°N	−5.6 °C	24.7 °C (July)	−26.0 / 40.0 °C	By the sea. 23 cm snowstorm on 7–8 Nov 2026

Place	Country	Latitude	January average	Hottest month average	Lowest / highest ever	What it means
Dalian	China	38.9°N	−3.3 °C	24.8 °C (August)	−21.1 / 37.3 °C	Mild, wet summers. Long heating season
Zhuanghe, Donggang (Liaodong coast)	China	39.7–39.9°N	−6.7 to −6.9 °C	about 27 °C average daily high in July	−28.1 / 37.7 °C	Coldest Chinese sites. July–August humidity 84–88%
Pyongyang	North Korea	39.0°N	−5.4 °C	25.0 °C (August)	−26.5 / 37.9 °C	Trade bans (sanctions). Shown for climate only
Morioka	Japan	39.7°N	−1.6 °C	23.5 °C (August)	−20.6 / 37.2 °C	Heat about 5–6 months. 209 cm of snow a year
Akita	Japan	39.7°N	0.4 °C	25.0 °C (August)	−24.6 / 38.5 °C	273 cm of snow a year

Sources: Chinese values are CMA 1991–2020 averages, taken from the Wikipedia climate tables for Beijing, Tianjin, Langfang, Tangshan, Qinhuangdao, Dalian, Zhuanghe and Donggang. North Korean values are KMA averages from Wikipedia. Japanese values are from JMA 1991–2020 normals. The 2026 snowstorm is from 环球网.



CMA normals via Wikipedia (Beijing), KMA via Wikipedia (Pyongyang), JMA (Morioka: January, February, August and December only)

Design conditions for the farm. Plan for an outdoor low of −20 to −28 °C and a high of 40–42 °C. Beijing fell to −19.6 °C on 7 January 2021. This was its coldest since 1966. It reached 40.7–41.1 °C on 22–23 June 2023, a June record.

Site type	Winter air	Summer air	What the rooms need
Inland Hebei and Tianjin	Dry: 40–55% humidity December–March	Hot: July average daily high 31.6–32.1 °C, highest ever 40–41.7 °C	Add water to the air in winter. Heat pumps must also cool
Liaodong and Hebei coast	Colder: January average −5.6 to −6.9 °C	Milder but wet: 80–88% humidity in July–August	Bigger heating system. In summer, air conditioning, not wet-pad coolers
Iwate and Akita	2–3 m of snow a year	Cool: August average 23.5–25.0 °C	Roof strong enough for heavy snow. Shortest cooling season in the band

In Beijing, the July average daily high is 31.8 °C. At 55% humidity, a wet-pad cooler brings incoming air down to about 26 °C. At 70% humidity, it only reaches about 28 °C. So humid sites need real air conditioning (research agent's calculation). We modelled heat for an insulated dubia room at 30 °C near Beijing. It uses 72–145 kWh of electricity per m² of rearing floor a year. January uses four to nine times as much as July. If power stops at −10 °C, a room gets too cold within hours. So every farm in the band must have backup power.

Global trends and current patterns

The insect business is now in two parts. Large-scale insect protein has failed. Small, high-value markets (live food for reptiles, and Chinese medicine) keep growing. But their prices go up and down a lot. A new farm should work in one of these small markets. It should sell to many buyers. It should expect prices to fall after every price rise.

Commodity insects collapsed. Ynsect raised over USD 600 million, but it closed in December 2025. In Japan, Gryllus (founded 2019) made food from crickets. In 2023, people protested about crickets in school lunches, and its sales fell. It left pet food in January 2024. It stopped work in spring 2024. It went bankrupt in November 2024 with JPY 153.39 million of debt (Wikipedia). Insect protein feed costs about ten times as much as soybean feed.

China is the only country with cockroach farms at industrial scale, and they serve medicine.

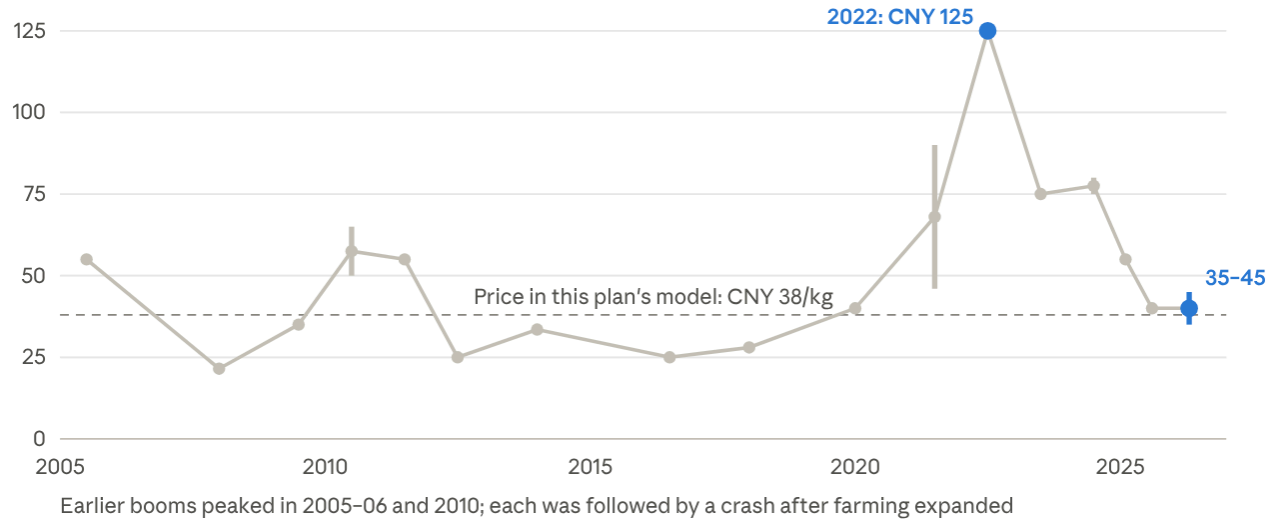
Maker	Where it gets American cockroach	Size	Source
Good Doctor (好医生), Xichang	Own farm, GAP quality approval 2009 and 2015	About 22 billion insects (15 billion phase 2 + 7 billion phase 1). About 28 t of feed a day. Harvest at 6–8 months	红星新闻, 2025
Kelun (科伦), Tengchong	Group farm 腾冲福德生物	150 t dried a year in 2019, plan of 300 t. 2025 request to buy feed for at least 300 t a year	云南风, 2019; Kelun, 2025
Jingxin (京新), Yunnan	Own farm, passed GAP quality check March 2024	Not stated	Company review (self-media)
Sainuo (赛诺), Dali	Company and farmers (old data)	100 kg dried gives 8–12 kg of extract	昆明日报, 2017

All four Kangfuxin makers use their own farms or group farms. We found no proof that any of them buys on the open market. Hospital sales of Kangfuxin fell from over CNY 2 billion (2022) to over CNY 1 billion (2024). This happened after group buying by hospitals cut prices by more than half (米内网 via 金融界). In Q1 2025 it still sold about CNY 340 million in public hospitals. It sold CNY 82.6 million in shops (retail pharmacies) (米内网).

Medicinal insect prices are cyclical and now low. Dried 土鳖虫 went from CNY 125/kg (2022) to CNY 35–45 (2026). Each price rise (2010, 2021–22) brought new farms. Then prices fell sharply. Dried American cockroach fell from CNY 50–55 (2022–24) to a CNY 29 weekly average in June 2026.

Dried 土鳖虫 fell from CNY 125/kg in 2022 to CNY 35–45 in 2026

Approximate market price, CNY per kg dried; bars show a stated range within the period



Approximate: 药通网 narrative price history via Sohu (2025-08-25), 2005 to August 2025; 2026 from 8u58 ungraded market reports

Reptile feeders are the growth market.

Measure	Value	Source
Small and exotic pet (小宠异宠) market, 2024	CNY 11.1 billion, up 10%+ a year	2025 小宠异宠白皮书 via 宠业家
Reptile and singing-insect (爬宠/鸣虫) online sales (GMV)	About CNY 2.4 billion, growing over 15% a year	Same
Share of those buyers who buy live feed	72%	Same
Spending of a repeat buyer compared with a new buyer	4.4 times	Same
Exotic-pet keepers in China	Over 17–19 million. Over 60% are 25–35 years old	澎湃, 2026; 2026 白皮书 summary (Baidu Baike)
Beijing reptile expo, 平谷, 19–21 June 2026	35,000 m², 650+ sellers, about 80,000 visits (seller report)	腾讯网; bilibili recap

Two warnings. First, the exotic-pet numbers cover all exotic pets, not only reptiles. Second, crested and gargoyle geckos can eat fruit-paste food (MRP) instead of live insects. This may reduce demand for live food (宠物投研院, self-media).

Hype and scams follow every boom. In 2026, officials said that “高价炒种、套路招商” (breeding insects sold at too high a price, and false buy-back offers) are common in 土元 farming (新华网山东). Dubia is now sold with “你养我收” buy-back offers. They promise “年纯利轻松十几万” from 50 m². Dubia is also sold under a new name, “黄金杜比亚”, as breeding insects at several times the market price.

Waste conversion lost to black soldier fly. In 2022–23, the Zhangqiu cockroach plant lost its kitchen waste to a black soldier fly plant. In 2025, a court sold its site. China's official list of animal feeds now includes black soldier fly, but no cockroach.

Markets, buyers and prices

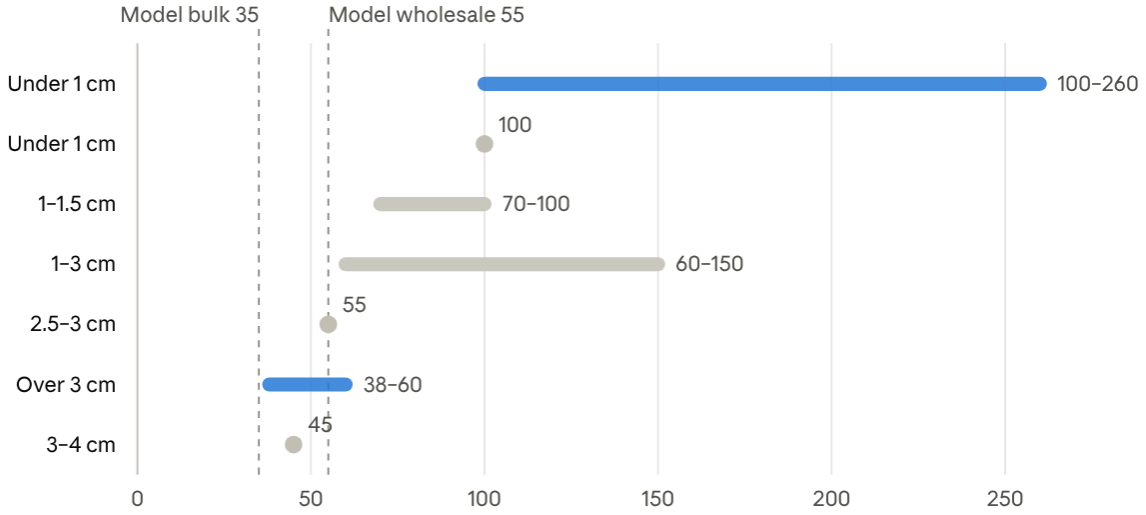
Wholesale buyers pay CNY 38–260 per jin (500 g) for live dubia. The price depends on size. This is three to ten times the price of mealworms. The Beijing–Tianjin area has buyers, but it has almost no local dubia farms. We could not open Taobao, JD or Pinduoduo. So every feeder price below has low reliability. These prices come from seller pages, Q&A sites and search results. The first job of the pilot is to replace them with real invoices.

Live feeders by size

Small young insects (nymphs) cost the most per jin. Adults cost the least. This is because one jin of small nymphs has thousands of insects. Sellers sort them by length: under 1, 1–1.5, 1.5–2, 2–2.5, 2.5–3 and over 3 cm.

Small nymphs fetch 2–4 times the adult price per jin

Live dubia wholesale, CNY per jin (500 g), by size; low-reliability seller and Q&A pages



Breeding groups are advertised at CNY 240–650 per jin. Model prices are CNY 110 and 70 per kg.

爱企查 user Q&A (2026-10-19) and a Zhihu seller post quoting July–August 2026 wholesale prices (2026-10-10); both low reliability

Size	Insects per jin (seller pages)	Wholesale price per 100 insects (our calculation)
Newborn, 0.5–1 cm	10,000–12,000	not calculated
1–1.5 cm	5,000–7,500	CNY 0.93–2.00
1.5–2 cm	1,500–2,300	CNY 2.6–10
Adult, 3.5–4 cm	50–70 (seems too low; weigh a sample)	CNY 64–90

The insect counts come from Baidu Wenku pages with no date. They are only estimates (Wenku). The research agent calculated the price per insect. In shops, packs of 100 insects sell for “几十元” (tens of yuan) (Sohu, 2025). This is 10–100 times the wholesale price per insect. Most of this cost is postage, packing and the promise that the insects arrive alive.

Other products and prices

Product	Buyers	2026 price	How reliable
Live dubia, shop price	Pet owners	CNY 200–300 per jin; “100 insects for tens of yuan”	Low (爱企查)
Live dubia, 1–1.5 cm, 1 kg pack	Resellers	CNY 220/kg (Hunan seller, sends by ZTO courier)	Low (1688)
Live dubia, large orders, at least 2,000 jin	Middlemen who collect from many farms	CNY 40 per jin (Fujian, Sept 2026 orders)	Low (seller post)
Breeding groups	New farms	CNY 240–650 per jin	Low; often sold with promises to buy back the output
Freeze-dried dubia	Pet shops, pet-food makers	about CNY 320/kg (Shandong sellers)	Low (1688)
Crickets	Pet owners	CNY 28.41 per jin	Medium (一亩田)

Product	Buyers	2026 price	How reliable
Mealworms (黄粉虫)	Pet owners, scorpion farms	about CNY 12–14 per jin; farms buy back at CNY 4–6 per jin	Medium
Dried 土鳖虫, not sorted by size	Anguo and Bozhou traders	CNY 35–45/kg; large and tested CNY 45–50	Medium (8u58)
Dried American cockroach	Traders; medicine factories and 饮片 (cut herbal medicine) factories	about CNY 29/kg on the open market; CNY 40–50 in two farm sales in 2026	Medium
Dubia for pet food, Japan	Pet owners	about JPY 1,000–2,500 per 100, by size	Estimate, not checked

In September 2026, Hong Kong customs took 13 kg of live dubia. Customs said they were worth HK$3,900. This means about HK$300/kg (南方都市报).

Pet-food insect buyers near Beijing and Tianjin

Buyer or competitor	Where	Distance from Beijing	Why it matters
十里河天娇文化城 and other pet markets	Chaoyang, Beijing	In the city	Reptile and cricket stalls; business is getting smaller
Beijing reptile expo and 平谷 reptile park	马坊, 平谷	about 50–60 km	Trade show every year, and a planned reptile business park
Gecko and lizard breeders	Tianjin, Liaoning	0–600 km	Buy feeder insects in large amounts; the Tianjin 爬房 (reptile house) welcomes visits
Dubia sellers, 阜城 (Hengshui)	Hebei	about 227 km	The only dubia sellers we found within 300 km: CNY 110–150 per jin
兴丰生态家庭农场 (example for comparison)	Suining, Jiangsu	outside our target area	70,000 jin of dubia sold in 2024; 1,000–2,000 parcels a day

Buyer or competitor	Where	Distance from Beijing	Why it matters
内蒙古峰子生物 (example for comparison)	Baotou	outside our target area	30 t of breeding insects; plans to double this in 2026

The research agent measured the distances in a straight line. A farm 1–2 hours from Beijing and Tianjin can deliver with its own van. Then it does not need couriers, which limit the sending of live animals.



A shop for reptiles and unusual pets (爬虫类宠物店) in a Shenzhen mall, February 2024. Shops like this sell feeder insects to pet owners. Photo: Rlwo MEUAD 20082, CC0 1.0, Wikimedia Commons.



A bearded dragon (a lizard) at the same shop. It has a dish of small young cockroaches. They look like dubia. Photo: RIwo MEUAD 20082, CC0 1.0, Wikimedia Commons.

Medicine buyers

China needs an estimated 2,000–5,000 t of 土鳖虫 a year (药通网 says 2,000 t; self-media says over 5,000 t). Caoxian in Shandong supplies about half. Buyers sort by size. They pay 25–30% more for lots that pass testing (包检).

Market	Date	Large, tested	Not sorted	Small
Bozhou	May 2026	CNY 45–46/kg	CNY 40	CNY 35
Bozhou	April 2026	CNY 50 (selected)	CNY 40–45	CNY 25–28
Anguo	January 2026	CNY 50 (selected)	CNY 45	CNY 25–28
Bozhou	January 2026	CNY 40	CNY 35	CNY 22
Anguo	November 2025	about CNY 40 (large)	CNY 30–35 (medium)	CNY 20–25

Source: 8u58 market reports, a website that collects prices. A page for shoppers gives CNY 35–45 per jin (CNY 70–90/kg) for Anguo lots that are not sorted. This is double the trade prices (百度健康). Plan the budget with the trade price. In 2026, 360+ households in Caoxian produced 900+ t of dried insects and 400+ t of egg cases. This was worth over CNY 120 million (新华网山东).



A wholesale shop in the Bozhou (亳州) herbal medicine market, April 2025. Bozhou and Anguo are China's two main herb markets. The price of dried 土鳖虫 is set there. Photo: Jeremy Thompson, CC BY 2.0, Flickr.



Cut pieces of cicada skin for medicine (蝉蜕饮片) at the Hua Tuo Memorial, Bozhou, June 2016. This is another medicine made from insects. Photo: 三猎, CC BY-SA 4.0, Wikimedia Commons.

Japan: a smaller market for feeder insects, with higher prices. There are pet shows in Tokyo and Yokohama. Send insects only inside Japan. We could not open any Japanese shop page. So the Japanese prices are not checked.

Country comparison	Northern China (Hebei)	Northern Japan (Iwate, Akita)	North Korea
Minimum wage	CNY 2,080–2,380 a month	JPY 1,090 an hour (Akita from 14 Oct 2026, Iwate from 1 Dec 2026)	Not applicable
Is a feeder-insect farm legal?	Legal. Under the Livestock Law, insects are not farm animals	Legal. Insect sellers do not need to register as animal dealers	UN resolutions 2375 and 2397 ban joint companies and the sale of machines to North Korea

Country comparison	Northern China (Hebei)	Northern Japan (Iwate, Akita)	North Korea
Average January temperature	−2.7 to −6.9 °C	−1.6 to 0.4 °C	−5.4 °C (Pyongyang)
Market	Large and growing market for unusual pets; few local dubia farms	Smaller, with higher prices; pet shows in Tokyo and Yokohama	No way to sell abroad
Our decision	Start here	Only a small farm with high prices, after we get local price offers	Do not operate

Sources: MOHRSS and Hebei notices; 岩手 and 秋田 labour bureaus; Wikipedia, Sanctions against North Korea. We could not check Japanese prices for electricity, rent and feeder insects.

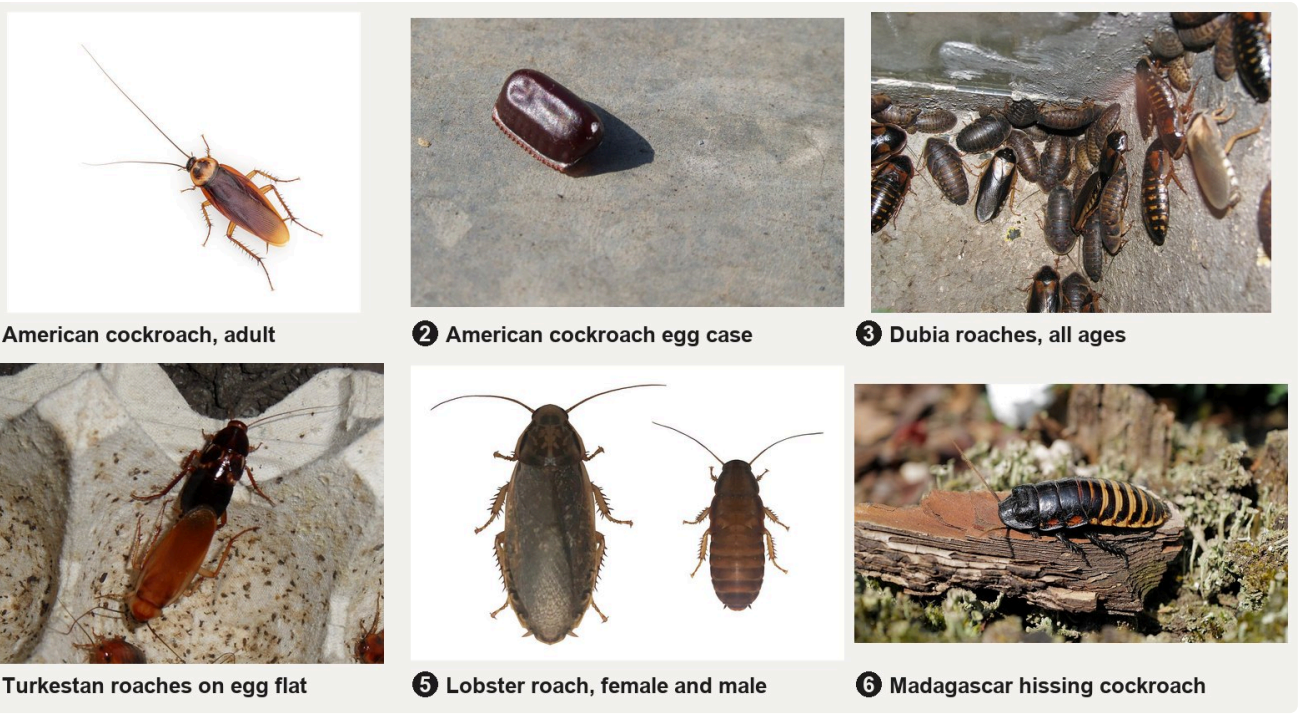
Breeding stock

Dubia is the best species to farm. The females give birth to live young. Dubia cannot climb smooth plastic. They are not on any Chinese or Japanese list of harmful foreign species. At 28–32 °C, one breeding female has about 10–35 young a month. The pilot needs about 3,000 females and 1,000 males (about 20 jin). We will buy them in stages for about CNY 6,000.

Species	Use in our target area	How it lives and breeds	Legal notes
Dubia (杜比亚, *Blaptica dubia*)	Live food for pets	20–35 live young per birth, one birth every 1–2 months. About 175 young per female a year in the best conditions. Adult in about 65–149 days at 30 °C. Females mostly stop breeding below 27 °C. Cannot climb smooth plastic	China: not on the 2022 main list of harmful foreign species. Not on the 2026 border control list. Never let them go free or throw them away. Japan: not on the list of harmful foreign species

Species	Use in our target area	How it lives and breeds	Legal notes
Turkestan roach (樱桃红, *Shelfordella lateralis*)	Small, fast live food	About 17 eggs per egg case. Adult in 126–279 days. Sold in small cups for CNY 2.4–8.8	Males can fly; keep them closed in carefully
Lobster roach (*Nauphoeta cinerea*)	Live food for pets	About 33 live young per birth	Can climb glass
Ground beetle cockroach (地鳖 *Eupolyphaga sinensis*; 冀地鳖 *Polyphaga plancyi*)	Dried females are sold as medicine	10–18 eggs per egg case; 8–10 hatch. Eggs hatch in 35–50 days at 27–32 °C. Ready to harvest in 14–15 months with heating in winter, or 2–3 years without heating. Females live 2–3 years. They lay an egg case about every 10 days	A local species in north China (冀地鳖 lives in Beijing, Tianjin and Hebei). Becomes inactive below 10 °C
Golden-edge cockroach (金边土鳖, *Opisthoplatia orientalis*)	Sold as high-quality 土元 in the south	A southern species (Fujian, Guangdong, Guangxi)	Not in the Pharmacopoeia (the official list of medicines); does not suit the north
American cockroach (美洲大蠊)	Medicine makers, only with a contract	4–6 months to adult at 28–30 °C. Stops moving at 10 °C and dies at 5 °C	A pest in cities; must be kept closed in very carefully

Sources: dubia biology from 爬宠窝 (hobby guide); 土鳖虫 from Baidu Baike and 药通网 via Sohu; invasive lists from gov.cn.



❶ American cockroach, adult

❷ American cockroach egg case

❸ Dubia roaches, all ages

❹ Turkestan roaches on egg flat

❺ Lobster roach, female and male

❻ Madagascar hissing cockroach

Photos via Wikimedia Commons: Insects Unlocked (CC0); Eran Finkle, Drägüs, Happy1892 (CC BY-SA 3.0); Ypna, Markbenecke (CC BY-SA 4.0).



Dried 土鱉虫 (*Eupolyphaga sinensis*), the product listed in the Pharmacopoeia. It is shown at the insect museum of National Taiwan University. Photo: Jerrythebest, CC BY-SA 4.0, Wikimedia Commons.



A live wingless female 地鳖 (*Eupolyphaga sinensis*). Only females are sold as medicine. Zhengzhou, Henan, 22 May 2026. Photo: Zinogre, CC BY-SA 4.0, iNaturalist.



Golden-edge cockroach (金边土鳖, *Opisthoplatia orientalis*), a southern species. It is sold as high-quality 土元, but it is not in the Pharmacopoeia. Shenzhen, July 2017. Photo: Donald Hobern, CC BY 2.0, Wikimedia Commons.

Numbers of insects for the pilot

Item	Value	Where the number comes from
Breeding insects at the start	about 3,000 females + 1,000 males (20 jin)	Our model; about 167 females per jin
Cost of the first insects	about CNY 6,000	Our model; bought in stages at market prices
Young per female a month	10–35 (about 15 on average)	Hobby guide, our calculation
Test that must be passed before the medium farm	at least 7.5 young insects for sale per female a month	This plan's test (half the average)
Males to females	1 male to 3–5 females (dubia); about 15% males (土鳖虫)	Hobby guide; 8u58

Item	Value	Where the number comes from
Breeding insects needed for each 1 jin of newborn young	at least 5 jin of adults (3 jin females + 2 jin males)	Common rule from self-media

- **Buy domestic stock only.** China does not allow people to bring or post live insects into the country without a health-check permit. This includes “昆虫类和其他无脊椎动物”. Buy from at least two well-known breeders. Visit their farms. Check for German cockroaches mixed in. Keep new insects apart for 30 days. Baotou's 峰子生物 says it has 30 t of breeding insects with no unwanted insects mixed in.
- **Pay market prices, never buy-back packages.** Sellers offer dubia breeding groups at CNY 240–650 per jin. They offer 土鳖虫 egg cases at CNY 35–70 per jin. A common trick is a promise to buy back the output at a high price. One seller in Ningyang offers “先养后付 包饲料 包技术 包回收”. For 土元, the usual trick promises to buy back at CNY 200+/kg. The real market price is CNY 40–50.
- **Avoid young partner firms.** A 2026 trade article says: be careful with any partner company registered for less than five years. Such companies often stop buying back, or they disappear (知乎, self-media).

Feed and nutrition

Feed and other supplies cost about CNY 14.3 per kg of live dubia in our model. This is about 10% of the average selling price. So feed cost does not decide if this business succeeds or fails. Hebei is a cheap place to buy feed. Its wheat bran has the lowest price of the four provinces that Mysteel reports.

Ingredient	Price in north China, early November 2026	Source
Corn	about CNY 2,224 per tonne (national, 6 Nov)	Mysteel
Wheat bran, price at the mill	CNY 1,351 (Hebei), 1,411 (Henan), 1,442 (Shandong) per tonne (4 Nov)	Mysteel
Soybean meal, 43%	about CNY 3,134 per tonne	Mysteel
Defatted rice bran meal	about CNY 1,556 per tonne	Mysteel
Fish meal, Peruvian	about CNY 26,200 per tonne; will stay high until the next fishing season	Mysteel

Model feed cost per kg of live dubia sold

Item	Quantity	Unit cost	CNY per kg live
Dry feed mix (40% bran, 30% corn, 20% soybean meal, a little fish meal and a vitamin-mineral mix called premix)	3 kg	CNY 3.6/kg	10.8
Vegetables and fruit (to give water)	1.5 kg	about CNY 1/kg	1.5
Egg trays, replacing worn boxes, gloves, cleaning	—	—	2.0
Total variable production cost			**14.3**

We assume 3 kg of dry feed per kg sold. This is a careful (high) number. It also covers feeding the breeding stock. In trials, dubia need a median of 1.7 kg of feed per kg of weight gain (Oonincx et al. 2015). At 1.2 kg/m², the pilot room uses about 216 kg of dry feed a month. The Suining farm that we use for comparison bought 480,000 jin of bran, 190,000 jin of corn and 30,000 jin of pumpkin in one year. This fed its dubia, mealworms and mice together. So the most feed it used is about 4.1 jin of feed per jin of insects (腾讯新闻).

- **Feeds named by dubia farmers:** wheat bran, corn meal, tofu waste (tofu dregs), mouse or dog food, vegetable leaves and fruit skins.
- **土鳖虫** eat bran, grain and green plants. One published mix for young nymphs is 45% bran, 3% soy, 2% animal protein and 50% greens (from social media). Our model assumes about CNY 16 of feed per kg dried.
- **Do not feed collected kitchen waste.** China does not allow it in animal farming. In 2026, 1,600 t of insect bait made from it was destroyed at a farm in Shiyan.

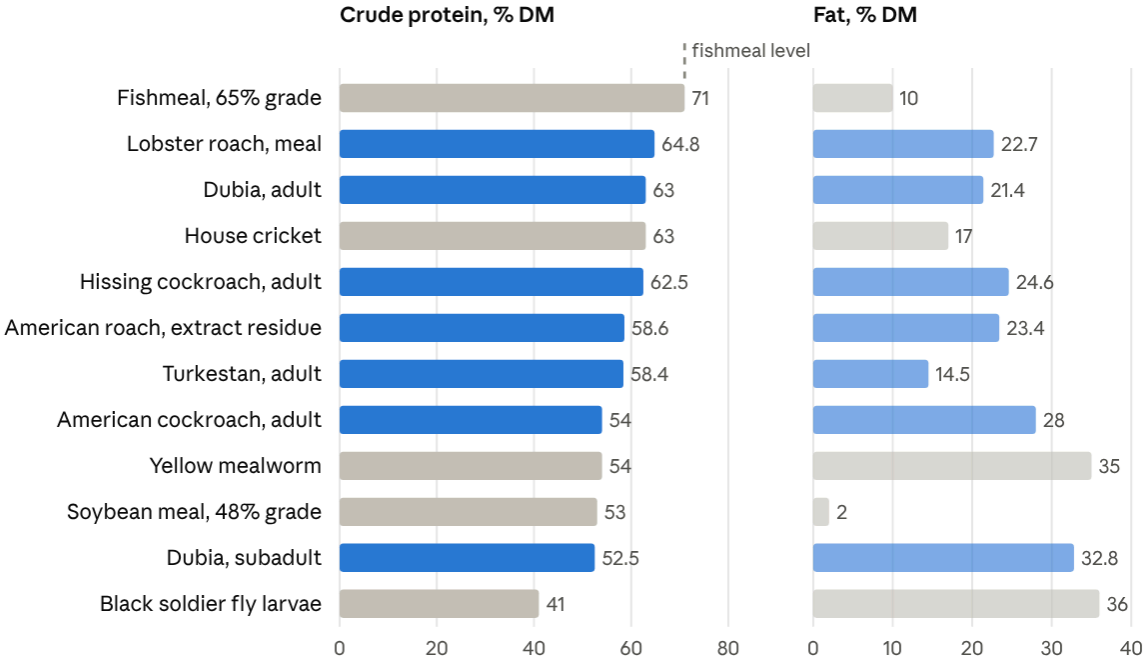
Nutrition and useful substances

Adult cockroaches are a food with high protein and medium fat. On a dry basis they have 58–65% crude protein and 14–25% fat. This is the same as crickets and more than soybean meal. But they are low in calcium, methionine and vitamins A and E. Their bigger value is in peptides, enzymes and other active substances (bioactives). In China, only licensed drug makers can sell these.

Protein and fat by species

Adult cockroaches carry 58–65% protein, 6–13 points below fishmeal

Crude protein and fat, % of dry matter · blue = cockroaches, grey = benchmark feeds



No published proximate figures found for 土鳖虫. Lobster roach measured as a dried meal; extract residue is what remains after ethanol extraction.

Kulma et al. 2016 (dubia, Turkestan), Oonincx and Dierenfeld 2012 (hissing cockroach), Fontes et al. 2019 (lobster roach meal), Bernard and Allen 1997 (American cockroach), 2026 black carp trial (extract residue), Feedipedia and Oonincx et al. 2015 (benchmarks)

Nymphs (young cockroaches) have more fat and less protein than adults. Subadult dubia have 33% fat and 52% protein. Adults have 21% fat and 63% protein (Kulma et al. 2016). A customer who wants protein should buy adults or large nymphs. A customer who wants energy for a thin reptile should buy small nymphs.

Water, chitin, minerals and vitamins

Species and stage	Water, % of fresh weight	Chitin (shell material) or fibre, % dry matter (DM)	Ash (minerals), % DM	Calcium (Ca), % DM	Phosphorus (P), % DM	Iron (Fe), mg/kg DM	Zinc (Zn), mg/kg DM	Source
Dubia, subadult	57.8	chitin 6.5	4.0	0.2	4.3*	—	—	Kulma 2016
Dubia, adult	62.6	chitin 6.4	4.9	0.2	4.7*	—	—	Kulma 2016
Turkestan, subadult	64.0	chitin 5.3	3.7	0.7	1.2	—	—	Kulma 2016
Turkestan, adult	69.3	chitin 5.9	5.1	0.9	2.5	—	—	Kulma 2016

Species and stage	Water, % of fresh weight	Chitin (shell material) or fibre, % dry matter (DM)	Ash (minerals), % DM	Calcium (Ca), % DM	Phosphorus (P), % DM	Iron (Fe), mg/kg DM	Zinc (Zn), mg/kg DM	Source
Turkestan, medium nymph	71.7	NDF 12.8	—	0.19	0.95	90	165	Oonincx and Dierenfeld 2012
Madagascar hissing cockroach, adult	61.1	NDF 34.6	—	0.10	0.55	111	216	Oonincx and Dierenfeld 2012
Lobster roach, dried meal	6.3	chitin 24.4	3.8	—	—	—	—	Fontes et al. 2019
American cockroach, adult	61	ADF 9	—	0.20	0.50	—	—	Bernard and Allen 1997
American cockroach, what is left after alcohol (ethanol) extraction	—	—	5.3	—	—	—	—	black carp trial, 2026
土鳖虫, dried (limits in the official drug standard, the Pharmacopoeia)	≤10	—	≤13	—	—	—	—	ChP 2025
House cricket	72	NDF 18	—	0.14–1.01	0.79	—	—	Feedipedia
Yellow mealworm	59	NDF 12	—	0.25	0.79	—	—	Feedipedia
Black soldier fly larvae, dried	8	NDF 17	—	3.30	0.72	—	—	Feedipedia
Fishmeal, 65% grade	8	—	14.9	4.48	2.86	—	—	Feedipedia; ash from the 2026 trial
Soybean meal, 48% grade	12	crude fibre 7	—	0.38	0.71	—	—	Feedipedia

*Kulma's phosphorus value for dubia (Ca:P 1:21.5–23.5) is much higher than other studies' 0.47–0.62%. It may be a measuring error. Chitin values depend on the test method. Fontes' 24% for lobster roach meal uses a different test from Kulma's 5–9%.

- **Calcium is the gap.** Every cockroach sample has a calcium-to-phosphorus ratio (Ca:P) below 1:1. Fishmeal has 1.5:1. Pet keepers cover the insects with calcium powder. Feed makers must add calcium.
- **Iron and zinc are high.** Turkestan and hissing cockroaches contain 90–111 mg/kg DM of iron and 165–216 mg/kg of zinc. The authors warn that too much iron is a problem for birds that easily get iron storage disease.
- **Vitamins are low.** Cockroaches contain 11–16 mg/kg DM of vitamin E and under 400 IU/kg of vitamin A. A cat-like meat-eating animal needs an estimated 5,000 IU/kg to stay healthy (Oonincx and Dierenfeld 2012). Turkestan nymphs were below the rat standard for five nutrients. Feeder insects usually lack vitamins A, D and E, thiamine, B12, calcium and iodine (Finke 2013).
- **土鳖虫 composition is published only by sex and stage, without figures we could open.** A 2022 study compared males and females. Mature females had more crude fat and extract. Males had more protein and amino acids. Males also had weaker anti-clotting (anticoagulant) activity. Adult males failed the Pharmacopoeia limits for ash and extract (Chang et al. 2022).

Amino acids

Cockroaches have a lot of lysine. Methionine is the amino acid in shortest supply (the limiting amino acid). Cockroaches have 0.7–1.4 g of methionine per 100 g of dry matter. Fishmeal has 1.8 g.

Amino acid, g per 100 g dry matter (DM)	Dubia, adult	Turkestan, adult	American cockroach (left after extraction)	Fishmeal
Lysine	6.3	9.4	3.5	5.0
Methionine	1.4	0.8	0.7	1.8
Cysteine	0.17	0.09	0.07	0.02
Threonine	1.7	2.4	2.4	2.6
Valine	6.6	8.1	3.5	3.2
Arginine	5.6	9.0	3.0	3.6
Histidine	3.7	4.8	1.6	2.0
Glycine	16.6	12.9	4.1	3.9
Essential amino acid score compared with egg white	0.7	0.6	—	—

Sources: Kulma et al. 2016 (dubia, Turkestan); the 2026 black carp trial (residue and fishmeal, same lab). Kulma's glycine and alanine values are unusually high. The authors think feed left in the gut caused this. Not feeding the insects before slaughter should lower these values. In the black carp trial, the diet had 20% residue. This lowered whole-body methionine and how well the fish used feed. So a feed made from the residue needs added methionine.

Fats. Oleic acid is the main fatty acid in dubia, Turkestan and *Blaberus* roaches. It is 38–44% of the total. Next is palmitic acid at about 22–27%. Linoleic acid is 9–18%, and all polyunsaturated fats together are 9.5–20.9% (Kulma 2016). Only Turkestan has more saturated fat than monounsaturated fat. In 土鳖虫, unsaturated fatty acids are about 75% of the fat that can be extracted. Linoleic acid is 28.5% of those (RSC Advances review, 2021).

Active substances for medicine

The value for medicine is in the parts that dissolve in water and in alcohol (ethanol). These are peptides, proteins that act as enzymes, polyols and small molecules. Most of the evidence comes from cell and animal studies. The exception is Kangfuxin Ye. It has many clinical trials, but they are small.

Substance	Species	What is known	Evidence	Source
Kangfuxin Ye: alcohol (ethanol) extract of the dried insect	American cockroach	Contains polyols, activity like epidermal growth factor, amino acids and mucopolysaccharide amino acids (words from the label). Taken by mouth for stomach and duodenal ulcers. Used on the skin for wounds, burns, fistulas and pressure sores. Quality control is based on amino acids. Good Doctor's patent requires total amino acids of at least 7.0% in the extract	A 2019 review of 22 randomised trials (meta-analysis, 2,024 patients) looked at adding it to standard stomach-acid drugs (proton-pump inhibitors). It improved the response of stomach ulcers (odds ratio 6.95). But the authors asked for better, stricter trials	Good Doctor label notice, 2021; Zou et al. 2019

Substance	Species	What is known	Evidence	Source
Other *P. americana* extracts and peptides	American cockroach	Slows tumours and the growth of new blood vessels (anti-angiogenic) in cells and mice. Licensed products include Xinmailong injection, Ganlong capsules and Xiaozheng Yigan tablets	Lab and animal studies (preclinical), plus small Chinese clinical reports	Zhao et al. 2017
Antimicrobial peptides (periplanetasins)	American cockroach	Periplanetasin-5 acts against cancer and inflammation. Periplanetasin-2 makes vancomycin and chloramphenicol work better against *E. coli*	Cell studies	Periplanetasin-5, 2021; periplanetasin-2, 2021
Proteins that dissolve blood clots (fibrinolytic: eupolytin1, EFP, EFF-1 to EFF-3)	土鳖虫	Dissolve fibrin (clot material) directly and activate plasminogen. Purified parts are 30–41 kDa. EFP stops new blood vessel growth and slowed S180 and H22 tumours in mice	Lab and animal	Yang et al. 2011; RSC Advances review
Alkaloids, nucleosides, unsaturated fatty acids	土鳖虫 (both Pharmacopoeia species)	21 alkaloids are reported, about 18 from *Polyphaga plancyi* (冀地鳖). In animal tests, total alkaloids widen blood vessels. Some kill cancer cells in the lab (cytotoxic)	Lab and animal	RSC Advances review
Chitin and chitosan	American cockroach	Chitosan output is 5.8% of the dried body. This is twice the German cockroach's 2.95% and more than mealworm's 1.7%. It kills bacteria in plate tests	Lab	2022 study

Substance	Species	What is known	Evidence	Source
"Milk" protein crystals	*Diploptera punctata* (not farmed here)	Proteins (glycosylated lipocalins) bound to fat, made in the embryo's gut. One crystal has over three times the energy of the same weight of cow's milk	Only basic structure research; no product	Banerjee et al. 2016

Allergens and safety limits. Cockroach tropomyosins (Bla g 7, Per a 7) are proteins that share about 82% of their structure with shrimp tropomyosin. So people with shrimp or dust-mite allergy may react to cockroach products. One Italian study looked at 21 children with severe shrimp allergy (anaphylaxis). All of them were sensitive to tropomyosins, and 76% to Bla g 7 (Miraglia del Giudice et al. 2023; Saetang et al. 2022). In January 2021, the NMPA changed the Kangfuxin Ye label. It added side effects and cases where the drug must not be used (Good Doctor notice above; we could not open the NMPA text). 土鳖虫 is listed as slightly toxic. The dose is 3–10 g, and it must not be used in pregnancy (ChP 2025). Products should carry an allergy warning like the one for shellfish. Workers should wear breathing masks (respirators).

Product catalogue

Product	Active substance or nutrient	Buyer	China	Japan	Price found	Maturity
Live dubia and Turkestan feeders	Whole-insect protein, fat and water	Reptile and amphibian keepers	Legal to breed and sell	Legal; dealers do not need to register	See Live feeders by size	On sale
Freeze-dried roaches	Same, at ⩽5% water	Pet shops, keepers	Not clear if a feed licence is needed	Pet Food Safety Act covers dogs and cats only	Fluker's US$13.64 per 1.3 oz, about US$370/kg in shops	On sale (mainly US)
Dried 土鳖虫	Proteins that dissolve clots, peptides, alkaloids	Bozhou and Anguo herb markets, makers of traditional Chinese medicine (TCM)	Medicine listed in the Pharmacopoeia (ChP 2025)	Very small imported market	Not sorted by grade: CNY 35.3/kg (weekly average, June 2026, 8u58)	On sale

Product	Active substance or nutrient	Buyer	China	Japan	Price found	Maturity
Patent medicines with 土鳖虫 (e.g. 大黄䗪虫丸, 跌打丸)	Whole dried insect used in the recipe	Hospitals, pharmacies	Licensed drugs; 34 recipes in the 2020 Pharmacopoeia (王晶 et al. 2022)	Not found	Not found	Licensed drug only
Dried American cockroach	Raw material for extracts	The four Kangfuxin Ye makers, wholesalers	Provincial standards (Sichuan, Yunnan)	Not found	CNY 29/kg weekly average, June 2026; price at the farm CNY 40–50/kg	On sale, mostly under contract
Kangfuxin Ye	Alcohol (ethanol) extract: polyols, activity like growth factor (EGF), amino acids, peptides	4,000+ hospitals, pharmacies	Licensed drug; over CNY 1 billion a year (The Paper)	Not approved (none found)	Retail price not found	Licensed drug only
Xinmailong injection (心脉隆)	*P. americana* extract	Hospitals	Licensed (Yunnan Tengyao); about CNY 298 million sales in 2017 (环球网)	Not found	Not found	Licensed drug only
Good Doctor products in development (美蠊复新液 and others)	Parts of the extract	Future hospital market	Phase II trial for mouth sores caused by radiation (mucositis) (ChiCTR2400084288)	—	—	Research
Cockroach powder as food	Protein	Online consumers	Not an approved food; sold without a standard	Not tried	About CNY 26 per jin	On sale illegally; avoid
Waste left after extraction, used as feed meal	58.6% protein, 23.4% fat (DM)	Feed mills, fish and poultry farms	Not in the official feed list; 2025–26 trials in carp, chickens, pigs and tilapia	Not found	Not found	Research
Chitosan	Chitosan, 5.8% of dried body	Buyers for cosmetics and medical materials	—	—	Not found	Research

Product	Active substance or nutrient	Buyer	China	Japan	Price found	Maturity
Antimicrobial peptides	Periplanetasins	Pharma research	—	—	—	Research
Frass (insect droppings)	Nitrogen, phosphorus, chitin	Gardeners, organic growers	Check if fertiliser registration is needed	Check	Black soldier fly frass €150/t in large amounts	A side product of every product line

For this farm, the product list confirms the order in the plan. First come live and freeze-dried feeders. Second is dried 土鳖虫 that meets the Pharmacopoeia standard. Extracts, meal and chitosan come only with a licensed partner.

Products and legal routes

Selling live dubia inside China is the clearest legal route. Insect farming counts as animal farming (code 0399). Cockroaches are not on any list of harmful foreign species. Insect farms do not need the farm registration or disease certificates that livestock farms need. Freeze-dried feed, frass (insect waste) and anything sold as food each need a separate check.

Product	China	Japan
Live feed insects and breeding stock	Legal to breed and sell. Releasing or throwing away foreign species: fine of CNY 10,000–50,000 (生物安全法). Bringing them in without approval: fine of CNY 50,000–250,000	Legal. Insects do not need animal-dealer (動物取扱業) registration (環境省)
Freeze-dried feed insects	Pet feed is taxed as feed. Check if a feed licence is needed. Cockroaches are not named in the feed catalogue (No. 1028, 2026)	The Pet Food Safety Act covers only dogs and cats (MAFF)

Product	China	Japan
Dried 土鳖虫	Official medicine list (Pharmacopoeia, 2025 edition). Limits: water ⩽10%, total ash ⩽13%, acid-insoluble ash ⩽5%, other matter ⩽5%, aflatoxin B1 (a mould poison) ⩽5 μg/kg, water-soluble extract ⩾22% (antpedia)	Very small market of imports. Not made in Japan.
Dried American cockroach	Only provincial standards (Sichuan SCYCBZXD2021-004: water ⩽9%, total ash ⩽8%, lead ⩽1 mg/kg). Sell only with a buyer contract.	—
Cockroach powder as food	Not an approved food. It is not on the new-food-ingredient list. It sells online at about CNY 26 per jin with no standard (The Paper). Do not sell it.	We will not sell it
Frass (insect waste) as fertiliser	Check if fertiliser registration is needed	Check

Regulatory checklist for a Hebei or Tianjin site

Rule	What it says	What it means for the farm
Industry classification	Census questions and answers: insect farming “暂定属于畜牧业” (for now counts as animal farming), code 0399 (同花顺)	The farm should get the cheaper farm electricity price. Confirm with the local 供电所 (power office).
Livestock Law (畜牧法)	畜禽 means the animals on the official breeds list. No insects are on it.	As written, the farm needs no 畜禽养殖场 filing. The no-livestock-zone (禁养区) rules do not apply.
Environmental impact	Only large livestock farms need a full report (名录 2021)	A small farm probably needs no environmental impact report (EIA). Ask the county office.

Rule	What it says	What it means for the farm
Farm buildings on farmland (自然资发〔2026〕146号)	Only listed uses are allowed. Each province decides what special farming is. Support buildings have a size limit (gov.cn)	Insects are not on any provincial list yet. Rent an existing building. Do not build on farmland.
Harmful foreign species	No cockroach on the 2022 list (59 species) or the 2026 port list (21 species)	Keep all insects closed in. Never throw away live insects.
Postal rules	邮政法实施细则 Art. 33 bans “各种活的动物” (all live animals) without special approval	Deliver near the farm in our own van. Send freeze-dried products to far buyers.

Tax in China. Farmers who sell their own farm products, including insects, pay no VAT (增值税法实施条例 Art. 26). Insects sold as pet food are taxed as feed (公告2026年第9号). Insect farms pay company income tax at half the normal rate. They do not pay zero (国家税务总局公告2011年第48号).

How the farm works

The process is simple. The hard part at 39–40°N is the climate. The rooms must stay at 29–32 °C and over 60% humidity. This must hold through a −20 °C winter and a 40 °C summer, with no long breakdown. Winter, not the yearly average, decides the size of the heat pumps and the generator.

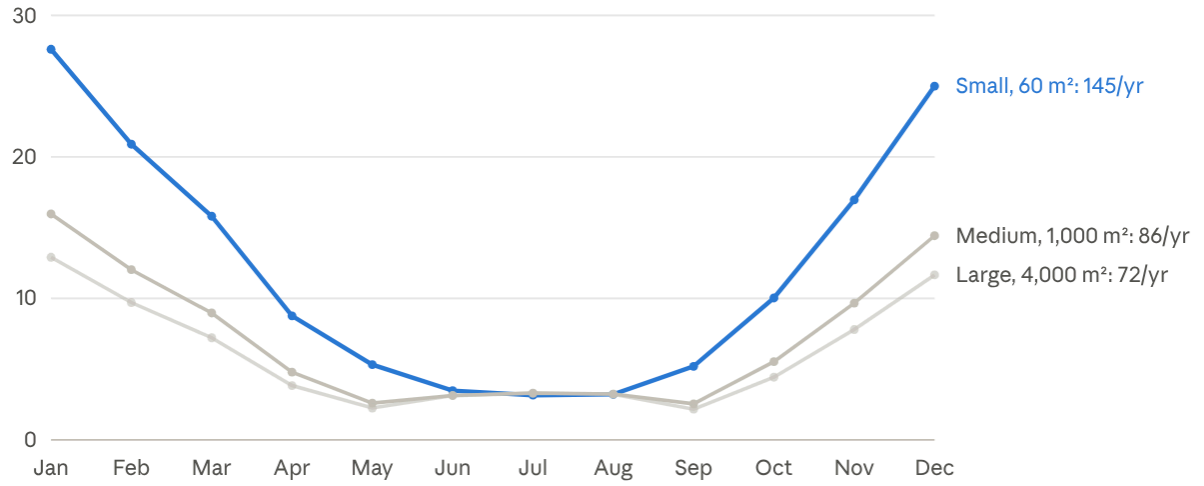
1. Quarantine room (a separate room to check for disease): 30 days for all new stock.
2. Breeding rooms: adults in smooth-walled boxes with egg trays, 1 male to 3–5 females.
3. Egg room (only for species that lay eggs: 土鳖虫 at 27–32 °C, 70–75% humidity).
4. Growing rooms: young insects (nymphs) are sorted by size every few weeks.
5. Cool sorting room with an air fan: sieving and sorting by size.
6. Processing: freeze-drying for feed insects. For 土鳖虫, boiling and then drying at about 50 °C.
7. Delivery: our own van within Beijing–Tianjin. In winter, parcels go with heat packs.

Climate targets. Keep dubia at 29–32 °C and 60–75% humidity. At 24–27 °C most females stop breeding. At 35 °C all young insects (nymphs) die. Keep 土鳖虫 at 25–30 °C. Their soil should “form a ball and fall apart” when squeezed.

Winter. Rooms of 100 mm insulated panels. Air-source heat pumps that work in deep cold. Humidifiers for the dry winter air. A generator big enough to run the heating. Coal is not possible. A notice from nine ministries set a goal to end loose coal on the northern plains by the end of 2025. This includes farm coal (农用散煤). Hebei has also banned coal in a zone next to Beijing since 2017.

January takes four to nine times the electricity of July

Heating, cooling and fans, kWh per m² of rearing floor per month, rooms at 30 °C near Beijing



Small room: 50 mm board retrofit, no heat recovery. Medium and large: 100 mm panels with heat-recovery ventilation.

This plan's heat-balance model (energy_north.py) on approximate Beijing 1991–2020 normals; heat-pump COP 1.8–3.5 by outdoor temperature; estimates

Heating item	Figure	Source
Most heat needed, insulated room	45 W/m² at −10 °C; 55 W/m² at −20 °C	Calculation by our research agent
1,000 m² hall	41 kW at −10 °C, 50 kW at −20 °C. About 55 MWh of electricity a season.	Same
Heat-pump efficiency (COP: heat out per unit of power in) at −10 °C	1.5–2.0 fixed-speed. 2.2–2.8 inverter (variable-speed).	Seller (芬尼科技)
Efficiency standard	GB 19577-2024 took the place of GB 37480-2019 from 1 Feb 2025	Industry news
3 HP home heat pump	CNY 9,000–18,000	Price survey by an online blogger
6 HP air-source heat pump	about CNY 9,800 per set	1688

Heating item	Figure	Source
25 HP unit for very low temperatures	CNY 50,000–80,000 installed	Short note from a seller
Diesel generator	10 kW CNY 3,588–3,950; 30 kW CNY 7,800	1688
Fuel cost compared with coal (coal = 100)	Biomass 65–85. Oil 380–400. Heat pumps cost more than coal and less than gas.	Heating seller

Summer. The same heat pumps also cool. Beijing has been hotter than 41 °C. Wet cooling pads help only in dry inland air.

Containment. Use smooth-walled boxes with a slippery band. Cover air holes with mesh. Use double doors (airlocks) and traps. Never throw away live insects. In 2013, about a million cockroaches escaped from a broken farm in Dafeng, Jiangsu (ETtoday).

Shipping. Postal rules limit the sending of live animals. Reports say SF Express stopped live-animal parcels from 1 December 2025 (online blogger, not confirmed). The average parcel in China takes 51 hours. About 13% take longer than 72 hours. Common heat packs last only 72 hours (国家邮政局). Sellers offer a “加热包 零下地区” (heat pack for below-zero areas) option. A foam box and 72-hour heat pack set costs about CNY 6–10. Use our own van within Beijing–Tianjin. Send freeze-dried products to far buyers.

Workers. Cockroaches cause allergies through the air. Keep air moving and give workers face masks (respirators).



1 Vibrating sieve grading by size



2 Rotary sifter removing frass



3 Freeze-dried insects, packed



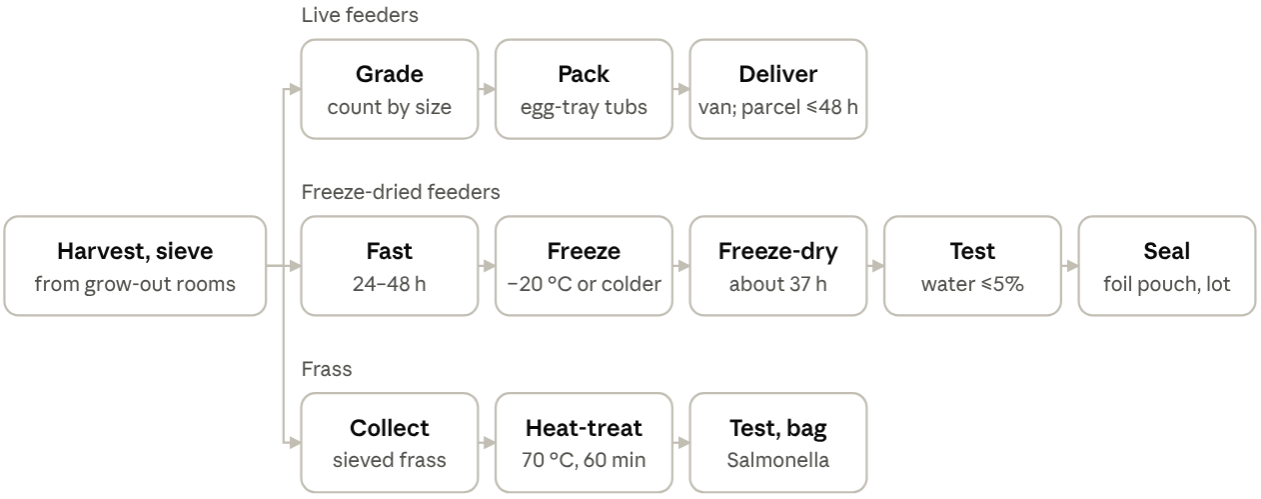
4 Insect frass for fertiliser

Stand-in photos of the same equipment at other insect farms: Stephen Ausmus (CC BY 2.0), Christophe Paul (public domain), Insektenwirtschaft.de (CC BY-SA 4.0), USDA-ARS (public domain).

Handling and processing

Every product leaves the farm by one of two lines. The feeder line sells insects alive or freeze-dried. The medicinal line kills, dries and tests each lot (batch). It tests to the national medicine book (Pharmacopoeia) or a provincial standard. The feeder line needs boxes, sieves and one freeze-dryer. The medicinal line needs a hot-water tank, a drying room and an outside laboratory. Figures marked *estimate* are this plan's working guesses. We will check them in the pilot (first small test farm).

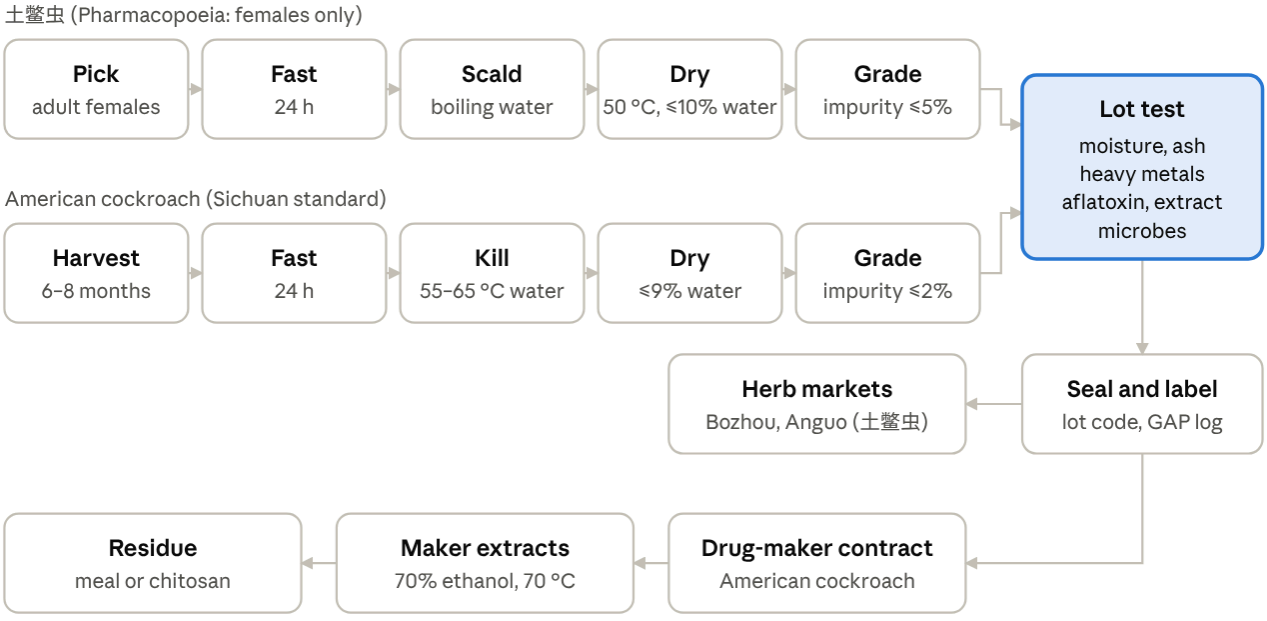
Feeder line: one harvest feeds three products



Feeder line · live, freeze-dried and frass branches from one harvest

One sieving step splits each harvest into three parts. The freeze-dryer takes about 37 hours a batch. It is the slowest step, so it sets how much the line can make.

Medicinal line: each lot is killed, dried and tested to its standard



Medicinal line · 土鳖虫 and American cockroach to sale or contract extraction

Both medicinal species follow the same five steps. Only the kill temperature and the moisture limit are different. We bag nothing for sale until its lot has passed the tests.

Moving live insects and breeding stock

Live insects die from heat, cold, crushing and their own moisture. They do not die from hunger. Pack them dry, with air holes and plenty of space. Keep the trip under 48 hours.

Item	What we do	Source
Container	A plastic tub with air holes covered by mesh. For large orders, a carton with double walls. In winter and summer, put a foam box around it.	DubiaRoaches.com ships in plastic tubs, and in boxes for large orders (FAQ)
Surface to climb on	Pieces of egg tray, standing up. The insects spread out and do not pile up.	Our own plan
How many per box	One layer of insects on each egg-tray surface. About 1,000 medium dubia in a 6–8 litre tub.	*Estimate*. We will test it with records of insects dead on arrival (DOA).
Food and water	A slice of carrot or potato, or water gel. No loose water, because it drowns young insects (nymphs) and makes cartons wet.	Our own plan
Temperature in the box	Aim for 15–30 °C. Josh's Frogs promises live dubia only from −18 to 38 °C by normal ground delivery. For next-day delivery the range is wider.	Josh's Frogs
Heat packs	Use 40–72-hour packs when it is below about 5 °C outside. Tape them to the lid, away from the insects. DubiaRoaches.com uses none, because packs can make a closed box too hot.	DubiaRoaches.com. Fluker's warns that heat builds up in closed boxes.
Cold packs	Only above about 30 °C, wrapped, with an air hole. In very hot weather it is better to stop shipping.	*Estimate*

Item	What we do	Source
Time on the road	48 hours or less. Ship only Monday to Wednesday. In China the average parcel takes 51 hours, and 13% take over 72.	国家邮政局
Claims for insects dead on arrival	The buyer sends a photo within 1 hour of delivery (DubiaRoaches.com's rule). We plan to replace 2–5% of insects.	Seller's rule. Our replacement figure is an *estimate*

Carrier rules differ sharply between China and Japan.

How it travels	Rule	What the farm does
China, postal and courier	邮政法实施细则 Art. 33 bans "各种活的动物" (all live animals) without special approval. The 2016 list of banned items does not name live insects (Wikisource). Reports say SF Express stopped live-animal parcels in December 2025. This is not confirmed.	Use our own van for Beijing–Tianjin. Send freeze-dried products to buyers far away. Get legal advice before sending any live parcels.
China, own vehicle	We found no rule just for insects	A van with heating and air conditioning. Tubs are tied into racks. A temperature recorder sits in the load space.
Japan, ゆうパック	Accepts insects that are healthy, harmless and have no smell. They must not need temperature control, air flow or food on the trip. They must not be able to escape. Japan Post does not pay if they die. Write the species and number on the label. If not, the parcel cannot go by air and is 1–4 days late (日本郵便)	Label "生きた昆虫（デュビア）約1,000匹" (live insects, dubia, about 1,000). Ship to Tohoku and Kanto only.

How it travels	Rule	What the farm does
Japan, Yamato and Sagawa	Yamato does not take animals. It takes only fish and insects that meet its rules. Sagawa may refuse animals (弁护士 JPニュース, 2024)	Use Japan Post as the normal choice
Moves inside the farm	Closed tubs go through the airlock (a double-door entry). Never open them in corridors. Each tub has a tag with room, box and date.	Our own plan. Rules to stop escapes are in Industrial process.
New breeding insects arriving	Keep them 30 days in the quarantine (separate) room. Count the dead on arrival. Check for mites, mould and parasites. Keep the supplier's batch apart until it breeds normally.	Our own plan

Storage

What is stored	Temperature	Humidity or moisture	How long	Notes
Live breeding stock	29–32 °C	60–75% RH (relative humidity)	No time limit	Same as in the breeding rooms
Live feeder insects kept for orders	24–27 °C. At this temperature female dubia mostly stop breeding.	50–60% RH	Up to 2 weeks (*estimate*)	Dry feed and water gel. Remove dead insects every day.
Insects that are starved or killed, before processing	−18 °C or colder	Sealed bags	Up to 3 months (*estimate*)	Freeze flat in thin layers, so they freeze fast

What is stored	Temperature	Humidity or moisture	How long	Notes
Freeze-dried feeder insects	Room temperature, dark	⩽5% moisture. Water activity (free water that lets germs grow) is about 0.07 after freeze-drying.	Best before 12 months (*estimate*)	Foil or mylar bag. Add an oxygen absorber, or fill with nitrogen gas.
Dried 土鳖虫	Cool	⩽10% moisture (ChP 2025)	Sell within one season. Prices change with the season.	"Ventilated, dry place, protected from insect damage" (ChP 2025). Aflatoxin B1 (a mould poison) ⩽5 µg/kg.
Dried American cockroach	Cool	⩽9% moisture (Sichuan standard)	As the buyer's contract says	Pesticide residues BHC and DDT must be ⩽0.2 mg/kg. So we never spray insect killer in the store.
Frass (insect droppings)	Dry	Below about 15% moisture (*estimate*)	12 months	Bag it after heat treatment. Keep it off the floor.

Pest control in stores. Dried insects attract mites, beetles, rats and mice. Use closed bins, or lined sacks on pallets. Use sticky traps and scent (pheromone) traps. Sell the oldest lot first. Do not spray, because the medicine standards limit pesticide residues.

Killing methods

First, starve the insects for 24–48 hours. Give them water but no feed. This empties the gut. It lowers the number of germs (microbial load). It also lowers feed left in the gut. Kulma's team said this leftover feed caused high glycine and alanine readings. Then choose the method for each product.

Method	How	Effect on quality	Animal welfare	Use for
Freezing	Cool to 4–8 °C until they stop moving. Then freeze at −18 to −40 °C.	Good for freeze-drying. In black soldier fly (BSF) larvae, the colour changed during freeze-drying.	Cooling first makes the insects slow before freezing	Freeze-dried feeder insects
Boiling water	Drop into boiling water	Kills germs on the surface. Makes the protein firm. This is the Pharmacopoeia method.	Death in seconds	土鳖虫 (置沸水中烫死, ChP 2025)
Hot water, 55–65 °C	Put in the water, rinse, and dry quickly	This is the Sichuan standard's method. Lower heat may protect peptides (small proteins) that heat can damage (*our reading*)	Death within about a minute (*estimate*)	American cockroach for medicine makers
Blanching (short boiling)	40 seconds in boiling water, then dry or freeze	Best of 10 methods in BSF larvae. The fat spoiled least (lowest lipid oxidation). It had fewer germs. It also starts the drying (Larouche et al. 2019)	Fast	Meal, oil, and freeze-dried treats that need fewer germs
Salt boiling	200 g salt per kg water, about 40 minutes	A heavier, salted product (盐水货). It is sold as a separate grade.	Fast	土鳖虫, only if a buyer asks
CO2 or nitrogen	Closed chamber filled with gas	Took 120–144 hours in BSF larvae. The fat spoiled more (Larouche 2019).	Slow	Not used
Dry heat, 60 °C	In an oven for 30 minutes	The fat spoiled more in BSF larvae	Slower than boiling	Not used

How we make each product

Live feeders. 1. Sort young insects (nymphs) by size through 3–4 sieves. 2. Count them by weight on a counting scale. Check one sample by hand in each batch. 3. Pack as above. Write species, size, number and lot on the label. 4. Deliver by van within 24 hours, or post Monday to Wednesday.

Freeze-dried feeders. 1. Starve 24–48 hours. 2. Freeze at −18 °C or colder. 3. Put them on trays in one layer. 4. Freeze-dry. Blanched mealworms took 2,235 minutes (37 hours) to freeze-dry. In a 70–90 °C oven they took 120–175 minutes. But freeze-dried ones kept a lighter colour and reached a water activity of 0.07 (Warsaw drying study, 2024). 5. Check that moisture is ⩽5% with an oven test. 6. Pack in foil bags with an oxygen absorber.

Dried 土鳖虫. 1. Pick adult females. Sell males separately. In one study, adult males failed the Pharmacopoeia limits for ash and extract. 2. Starve 24 hours. 3. Kill in boiling water, then rinse. 4. Dry in the sun or in an oven at about 50 °C. Stop when the belly has nothing soft inside. 5. Blow away and sieve out dirt until it is ⩽5% of the weight. 6. Test each lot against the ChP 2025 limits.

Dried American cockroach. 1. Harvest adults at 6–8 months. 2. Starve 24 hours. 3. Kill in 55–65 °C water and rinse. Dry quickly to ⩽9% moisture. 4. Sieve until dirt is ⩽2%. 5. Test against the Sichuan standard and the buyer's contract.



❶ Insects in boiling water



❷ Hot-air tray dryer



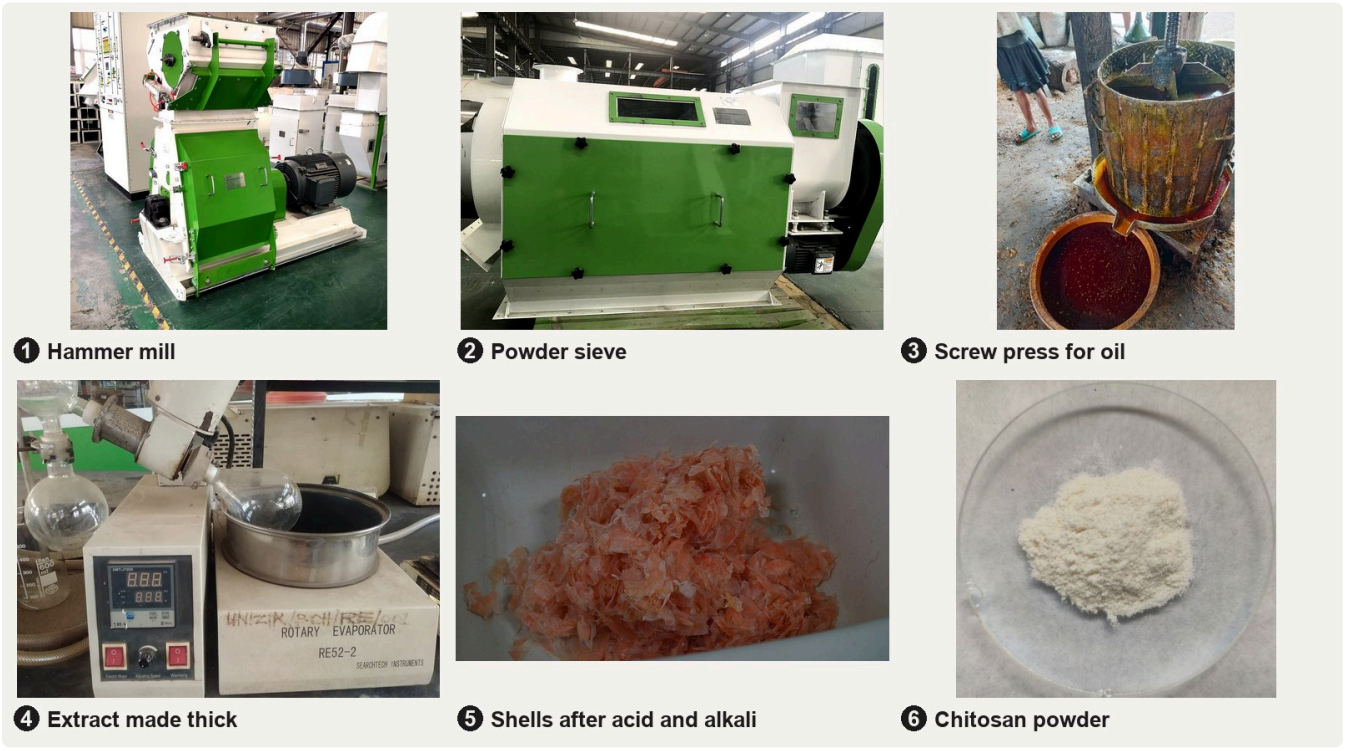
❸ Freeze-dryer trays

Killing and drying. These are stand-in photos. They show the same steps with other foods, not with cockroaches. 1. Silkworm pupae in boiling water, the same as our kill step in hot water. Photo: Charles Haynes, CC BY-SA 2.0. 2. A hot-air dryer with trays. Photo: Waz8, CC0. 3. Trays come out of a freeze-dryer. Photo: USDA-ARS, public domain. All from Wikimedia Commons.

Powder or meal. 1. Blanch and dry. Or use what is left after extraction. 2. Press out the oil in a screw press. Or remove the fat with a solvent at a licensed plant. 3. Grind in a hammer mill and sieve to 40–60 mesh. 4. Test protein, fat, moisture and germs. In China, sell it only as animal feed, never as food.

Extract (licensed maker only). Good Doctor's patent uses 1,000 kg of crushed dried insects. It soaks them in 4 volumes of 70% ethanol at 70 °C for 8 hours. Then it uses 3 volumes for 6 hours. Then it makes the liquid stronger and removes the fat (US10155013B2). Sainuo reported 8–12 kg of active extract from 100 kg of dried insects (昆明日报 via 搜狐). The farm's work ends with the dried, tested insect.

Chitin and chitosan (research). Remove minerals in weak acid. Remove protein in weak alkali. Then treat in 50% sodium hydroxide at 100 °C for 4 hours, three times (deacetylation). This gave 5.8% chitosan from dried American cockroaches (2022 study).



Meal, oil, extract and chitin. These are stand-in photos from other industries. 1. A hammer mill for feed. Photo: RICHI Machinery, CC BY-SA 4.0. 2. A sieve for powder. Photo: RICHI Manufacture, CC BY-SA 4.0. 3. A simple screw press makes palm oil. Photo: Joy Fubi, CC BY 4.0. 4. A rotary evaporator takes the liquid out of an extract, so it gets thick. Photo: Adimora chidinma, CC0. 5. Shrimp shells after the acid and alkali steps. Insect shells go through the same steps. Photo: Serhii Tsykunkov, CC0. 6. Chitosan powder. Photo: Aariuser I, CC BY-SA 2.0. All from Wikimedia Commons.

Frass. Sieve it from the bins. Heat it at 70 °C for 60 minutes. This is the EU rule, and a good rule for other places too. Then cool, test and bag it.

Product line	What goes in	Main equipment	Output (yield)	Time and amount per batch	Energy	Tests before sale
Live feeder insects	Young insects sorted by size	Sieves, counting scale, tubs	About 1:1, minus 2–5% dead on arrival (*estimate*)	Packing every day	Very little	Number, size, share dead on arrival
Freeze-dried dubia	Adults, starved and frozen	A large freeze-dryer of home size. It takes 8–12 kg fresh a batch, up to about 2,300 kg a year (Harvest Right)	0.39 kg per kg live for adult dubia (Kulma found 37.4% dry matter. We dry to 5% water. *our calculation*)	About 37 hours a batch	About 1.5 kW on average. That is about 55 kWh a batch, or 5–7 kWh per kg fresh (*estimate*)	Moisture ⩽5%. No Salmonella in 25 g. Enterobacteriaceae (gut germs) ⩽300 CFU/g. These are EU feed limits.
Dried 土鳖虫	Live females	Hot-water tank, tray dryer at 50 °C	0.32–0.43 kg dry per kg fresh (Baidu Baike and growers). Second-year females give about 30% less.	One drying day per batch (*estimate*)	About 0.6–0.9 kWh per kg fresh for a hot-air dryer (*estimate*)	ChP 2025: water ⩽10%, total ash ⩽13%, ash that does not dissolve in acid ⩽5%, aflatoxin B1 ⩽5 µg/kg, extract that dissolves in water ⩾22%
Dried American cockroach	Live adults	Hot-water tank at 55–65 °C, dryer	About 0.43 kg per kg live (61% water, dried to 9%. *our calculation*)	One day per batch (*estimate*)	As above	Sichuan: water ⩽9%, total ash ⩽8%. Metals: Pb (lead) ⩽1, Cd (cadmium) ⩽0.3, As (arsenic) ⩽1, Hg (mercury) ⩽0.2, Cu (copper) ⩽20 mg/kg. Alcohol extract ⩾8%. Total amino acids ⩾0.45%.
Meal and oil	Dried insects	Screw press, hammer mill, sieve	Oil is about 15–20% of dry weight when fat is 21–28% (*estimate*). The rest is meal.	By batch	Small	Protein, fat, moisture, Salmonella

Product line	What goes in	Main equipment	Output (yield)	Time and amount per batch	Energy	Tests before sale
Extract	Dried American cockroach	The licensed medicine maker's plant	8–12 kg per 100 kg dried	14 hours of extraction	Paid by the maker	Total amino acids ≥7.0% (patent)
Chitosan	Dried insects or residue	Glass-lined tanks for chemical reactions	5.8% of dried body	About 3 days (*estimate*)	High	Degree of deacetylation (how pure the chitosan is)
Frass	Sieved droppings	Sieve, oven or kiln at 70 °C	Side product	60 minutes at 70 °C	Small	No Salmonella in 25 g. E. coli or Enterococcaceae ≤1,000 CFU/g (EU).



❶ Oven for the water test



❷ Furnaces for the ash test



❸ HPLC lab, herbal maker

Tests before sale. These are stand-in photos from other labs. 1. A drying oven. It measures the water in a sample. Photo: Karelj, public domain. 2. Furnaces burn a sample to measure its ash. Photo: Brammers, CC0. 3. The HPLC lab of a herbal medicine maker in France. HPLC measures the active substances. Photo: Root66, CC BY-SA 3.0. All from Wikimedia Commons.

Traceability. Give every lot a code. The code shows room, box, harvest date and process date. Record which feed batch each lot ate. Medicine buyers expect records in the style of GAP (Good Agricultural Practice). They also expect a test report (certificate of analysis) for each lot, and kept samples. An outside laboratory tests for heavy metals, aflatoxin and germs. This is a running cost in the plan. Feeder buyers in Japan mainly need the species, the number and a record of safe arrival.

Farm set-ups by size

There are three dubia set-ups. The largest has about 70 times the floor area of the smallest. All three use one design: insulated panel rooms, heat pumps with a spare, a generator, and smooth plastic boxes on steel shelves. We show output at two levels: the

careful case and the better case. Output (kg per m² a month) is the one number that decides if the larger farms make money.

	Small	Medium	Large
Building	60 m² insulated room in an existing building	1,000 m² of growing rooms in a rented 1,300 m² rural building	4,000 m² in a rented 5,000 m² building
Output produced at 1.2 kg/m² a month	about 72 kg of live dubia a month	about 1.2 t a month	about 4.8 t a month
Output produced at 2.0 kg/m² a month	about 120 kg a month	about 2 t a month	about 8 t a month
Share sold	92%	95%	95%
How sales are split (retail / wholesale / bulk / freeze-dried)	25 / 50 / 25 / 0%	15 / 45 / 25 / 15%	5 / 35 / 35 / 25%
Average price before costs	CNY 147.5/kg	CNY 123.8/kg	CNY 97.7/kg
Equipment	2 inverter air conditioners that also heat, oil heater as backup, dehumidifier, sensors with phone alarm, 10 kW generator, 36 shelves, 360 boxes, 3,600 egg trays	6 × 6 HP heat pumps (one is a spare), air system that keeps heat, 50 kW generator, 600 shelves, 6,000 boxes, 3 shaking sieves, freeze-dryer, van	21 × 6 HP heat pumps, 150 kW generator, 2,400 shelves, 24,000 boxes, machine sieving, freeze-drying line, two vans
Electricity for heating and cooling	145 kWh/m² a year	86 kWh/m² a year	72 kWh/m² a year
People	Owner, full-time	12 staff (8 workers, 2 packers, 1 sales, 1 manager)	about 35 staff (24 workers, 5 packers, 3 sales, 3 managers)
First sales	month 4	month 5	month 6
Time to full output	about 15 months	about 26 months	about 32 months

Model inputs (model_cn.py with northern Hebei costs). The average price falls as the farm gets bigger. Larger farms must sell more in bulk and freeze-dried, and these prices are lower.

Evidence on yield. We found only one real farm figure, from Suining. It made 8,000 jin of dubia every three months from 1,500 m². It shared that space with mealworms and mice. That is at least 0.9 kg/m² a month (韩城新闻网, copied report). Online bloggers give simple rules: 500 boxes per 100 m² and about 3 kg per box a year (1.25 kg/m² a month). They also give 10–15 kg/m² of insects in the rooms at any time. So 1.0–1.2 kg/m² is the realistic base. For this reason the plan treats 2.0 as a better case that we must still prove.



Stand-in photos from cricket and fruit-fly farms: Brian Gratwicke and Richard Allaway (CC BY 2.0), Christophe Paul (public domain).

Investment and financials

We use a careful base output of 1.2 kg of live dubia per m² a month. At this output, only the small farm makes money. It earns back its money in month 24. The medium and large farms only just cover their operating costs (EBITDA). They never earn back their start-up money.

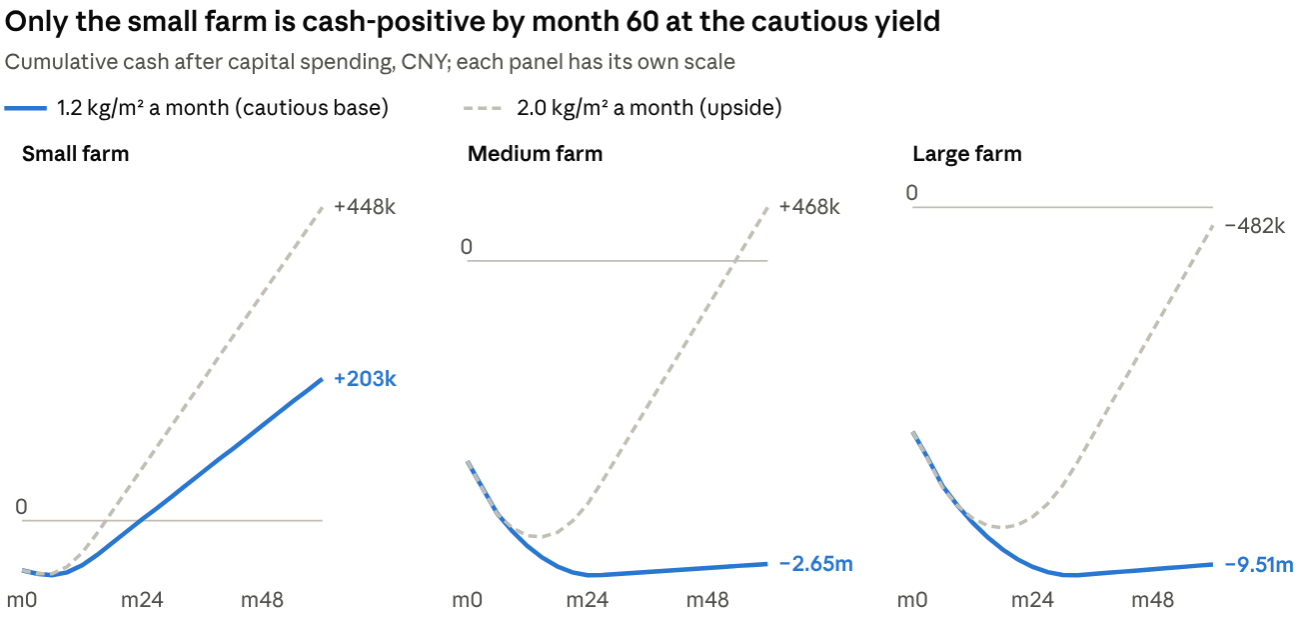
At 2.0 kg/m², the medium farm earns back its money in month 54 and the large farm in month 63. So the pilot must prove the output before anyone invests more.

Basis. Our model runs month by month for ten years. It shows results before tax, depreciation and loans. It uses northern Hebei costs. Power costs CNY 0.55/kWh. A worker costs CNY 4,500 a month, plus about CNY 1,100 of employer social insurance. Rent is CNY 8 per m² a month. Electricity for heating and cooling comes from our heat model. The dubia price is an average across sales channels: retail CNY 300/kg, wholesale CNY 110/kg, bulk CNY 70/kg, and freeze-dried CNY 260/kg dried. These prices are not very reliable, so we must test them in the first year. Scripts: model_cn.py (base), model_cn_stress.py (northern costs) and model_inv.py (this plan's tables).

Summary by scale

	Small dubia, 60 m²	Medium dubia, 1,000 m²	Large dubia, 4,000 m²
Start-up cost (capex)	CNY 70,840	CNY 1,724,000 + CNY 31,200 deposit	CNY 5,854,000 + CNY 120,000 deposit
Start-up cost in USD	USD 10,600	USD 263,000	USD 896,000
Sales a year at full output, 1.2 / 2.0 kg	CNY 117,000 / 195,000	CNY 1.69 / 2.82 million	CNY 5.35 / 8.91 million
Profit before tax, interest and depreciation (EBITDA) a year at full output, 1.2 / 2.0 kg	CNY 67,000 / 124,000 (owner not paid)	CNY 35,000 / 871,000	CNY 126,000 / 2.80 million
Time to earn back the money, 1.2 / 2.0 kg	month 24 / month 17	never / month 54	never / month 63
Most money needed at one time, 1.2 / 2.0 kg	CNY 78,700 / 78,400	over CNY 2.75 million* / CNY 2.42 million	over CNY 9.8 million* / CNY 8.54 million
Total cash at month 60, 1.2 / 2.0 kg	+CNY 203,000 / +448,000	−CNY 2.65 million / +468,000	−CNY 9.51 million / −482,000

*These cases never pay back within ten years. The figure shows the lowest total cash point. An investor would have to pay this with no sign of getting it back.



Month 0 = capital spent (plus rent deposit). Small farm with the owner unpaid. Model estimates, northern Hebei costs.

This plan's model (model_inv.py on model_cn.py, northern Hebei costs), quarterly points, months 0–60

Start-up cost by item

Small farm (60 m²)

Item	CNY
Insulate and seal an existing 60 m² room (boards, panels, sealed door, smooth band on the walls)	15,000
Two 1.5 HP inverter air conditioners that also heat, an oil heater as backup, a dehumidifier	7,600
Thermostats, 3 sensors with phone alarm	600
10 kW diesel generator	3,800
36 five-tier steel shelves	9,000
360 storage boxes with mesh vents	9,000
3,600 egg trays	1,100
Sieves, small shaking screen, scale, chest freezer, packing table	5,500
Breeding insects, bought in stages: about 3,000 females + 1,000 males (20 jin)	6,000
Online shop set-up, deposit, photos	3,000

Item	CNY
Business licence, contract checks	1,000
Extra for surprises (15%)	9,240
Total	**70,840**

Medium farm (1,000 m²)

Item	CNY
Building work: insulated panel rooms, sealed floors, double-door entries, drains (CNY 450/m²)	450,000
6 × 6 HP air-source heat pumps, with 1 spare; fresh-air system that keeps heat; humidity control	140,000
Better electrical supply, LED lights, 50 kW diesel generator that starts by itself	80,000
600 five-tier shelves, 6,000 boxes, 60,000 egg trays	330,000
Sensors, alarms, cameras	15,000
3 shaking screens, scales, cool workroom with air outlet	40,000
Freeze-dryer, 5–10 m² (our estimate; we found no shop price)	200,000
Packing and dispatch area	30,000
Used van	60,000
Breeding insects, bought in stages: about 30,000 females + 10,000 males (200 jin)	56,000
Brand, online shops, packaging design	30,000
Licences, land and lease checks, plan to keep pests and disease out (biosecurity)	20,000
Extra for surprises: 20% of building and equipment	273,000
Total (plus CNY 31,200 rent deposit)	**1,724,000**

Large farm (4,000 m²)

Item	CNY
Building work (CNY 350/m²)	1,400,000
Heating and cooling: 21 × 6 HP heat pumps with spares, fresh-air system that keeps heat, humidity control	470,000
Better electrical supply and 150 kW generator	300,000
2,400 shelves, 24,000 boxes, egg trays	1,100,000
Sensors, alarms, cameras, business software (ERP) and order system	100,000
Machine lines to sieve and sort by size, scales, cool workrooms	300,000
Freeze-drying line, about 20 m² (our estimate)	600,000
Packing hall and cold store	150,000
Two vans	150,000
Breeding insects, bought in stages (750 jin)	190,000
Brand, sales channels, packaging; design, licences, land checks, biosecurity	160,000
Extra for surprises: 20% of building and equipment	934,000
Total (plus CNY 120,000 rent deposit)	**5,854,000**

We checked equipment prices against 1688 listings where we could. A 6 HP heat pump costs about CNY 9,800. A 10 kW generator costs CNY 3,588–3,950. Egg trays cost CNY 0.26–0.30 each. Crates cost CNY 4.4–25. We checked building costs against seller figures. A complete steel building costs CNY 600–1,000/m². Sandwich panel costs CNY 150–250/m². If we rent an existing building, the building work stays at CNY 350–450/m².

Monthly running costs at full output (1.2 kg/m²)

CNY a month	Small	Medium	Large
Selling costs (parcels, platform fees, insects that arrive dead)	1,590	19,512	43,137
Feed and supplies (CNY 14.3/kg produced)	1,030	17,160	68,640

CNY a month	Small	Medium	Large
Staff, with social insurance	0 (owner not paid)	71,200	214,000
Electricity for heating and cooling, yearly average	399	3,942	13,200
Fixed costs (rent, repairs, marketing, insurance, office work, reserve)	1,150	26,400	96,000
Total operating cost	**4,169**	**138,214**	**434,977**
Sales	9,770	141,153	445,498
EBITDA	**5,602**	**2,939**	**10,521**

Staff are half of the medium farm's costs. This is why its result depends so much on output. The same 12 people cost the same at 1.2 or 2.0 kg/m². Electricity is under 5% of costs on average. But January use is 2.2–2.3 times the average. So the winter peak decides the size of equipment and backup. Medium fixed costs include CNY 10,400 of rent for 1,300 m².

Unit economics per kg of live dubia sold

CNY per kg sold	Small, 1.2 kg	Medium, 1.2 kg	Medium, 2.0 kg
Average price	147.5	123.8	123.8
Selling costs	24.0	17.1	17.1
Feed and supplies	15.5	15.1	15.1
Staff	0	62.5	37.5
Electricity	6.0	3.5	2.1
Fixed costs	17.4	23.2	13.9
EBITDA per kg	**84.6**	**2.6**	**38.2**

We calculated this from the monthly costs above. At medium size, going from 1.2 to 2.0 kg/m² adds about CNY 36 of operating profit (EBITDA) per kg. Almost all of it comes from spreading staff and fixed costs over more output. An investor is buying this effect. So we must measure output first.

Five-year profit and loss

Small farm, 1.2 kg/m² (owner unpaid)

CNY	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	36,639	114,802	117,245	117,245	117,245
Selling costs	5,962	18,680	19,077	19,077	19,077
Feed and supplies	4,393	12,098	12,355	12,355	12,355
Electricity	3,613	4,745	4,785	4,785	4,785
Fixed costs	13,800	13,800	13,800	13,800	13,800
EBITDA	**8,872**	**65,480**	**67,227**	**67,227**	**67,227**
Owner's wage, if taken (CNY 5,600 a month)	67,200	67,200	67,200	67,200	67,200

At 2.0 kg/m², the small farm's operating profit (EBITDA) is CNY 26,395 in year 1. From year 2 it is CNY 121,496–124,436 a year.

Medium farm, 2.0 kg/m² (upside) and 1.2 kg/m² (cautious)

CNY	Year 1	Year 2	Year 3	Year 4	Year 5
Sales, 2.0 kg	384,962	1,860,650	2,812,361	2,823,055	2,823,055
Selling costs	53,215	257,205	388,765	390,243	390,243
Feed and supplies	69,030	226,200	341,900	343,200	343,200
Staff	503,384	694,200	852,620	854,400	854,400
Electricity	29,634	39,238	47,210	47,300	47,300
Fixed costs, with rent	316,800	316,800	316,800	316,800	316,800
EBITDA, 2.0 kg	**−587,101**	**327,007**	**865,066**	**871,112**	**871,112**
Sales, 1.2 kg	230,977	1,116,390	1,687,417	1,693,833	1,693,833
EBITDA, 1.2 kg	**−692,188**	**−223,891**	**32,388**	**35,267**	**35,267**

Large farm

CNY	Year 1	Year 2	Year 3	Year 4	Year 5
Sales, 2.0 kg	769,997	4,454,982	8,139,967	8,909,964	8,909,964
EBITDA, 2.0 kg	−2,034,914	333,013	2,370,320	2,796,025	2,796,025
Sales, 1.2 kg	461,998	2,672,989	4,883,980	5,345,978	5,345,978
EBITDA, 1.2 kg	−2,220,807	−1,001,872	−68,730	126,255	126,255

Five-year cash flow

Cash from operations is operating profit (EBITDA) minus the money tied up in stock and in unpaid customer bills. The medium farm lets 40% of customers pay after 30 days. The large farm lets 60% pay after 45 days.

CNY	Year 0 (start-up cost + deposit)	Year 1	Year 2	Year 3	Year 4	Year 5	Cash at end of year 5
Small, 1.2 kg	−70,840	6,872	65,480	67,227	67,227	67,227	+203,194
Small, 2.0 kg	−70,840	24,395	121,496	124,436	124,436	124,436	+448,358
Medium, 1.2 kg	−1,755,200	−742,719	−254,688	27,255	35,267	35,267	−2,654,818
Medium, 2.0 kg	−1,755,200	−651,320	275,678	856,511	871,112	871,112	+467,894
Large, 1.2 kg	−5,974,000	−2,424,756	−1,180,072	−187,529	126,255	126,255	−9,513,847
Large, 2.0 kg	−5,974,000	−2,308,163	36,014	2,172,320	2,796,025	2,796,025	−481,778

Lowest prices that still work

Average price before costs, CNY per kg live	Small, 1.2 / 2.0 kg	Medium, 1.2 / 2.0 kg	Large, 1.2 / 2.0 kg
Price in our model	147.5	123.8	97.7
Price where operating profit (EBITDA) is zero	53.0 / 42.5 (owner not paid); 147.5 / 99.2 (owner paid)	121.0 / 82.0	95.3 / 65.2
Price needed to earn back the money in 5 years	80.9 / 59.4 (owner not paid)	180.4 / 117.8	151.5 / 99.3

At 1.2 kg/m², the medium and large farms need prices within 2–3% of today's average price. This only covers cash costs. For both farms, operating profit (EBITDA) turns positive at about 1.17 kg/m². To earn back the money in five years, the medium farm needs about 1.9 kg/m². The large farm needs about 2.0 kg/m².

What if: changes in price and output

Case	Small, 1.2 kg	Small, 2.0 kg	Medium, 1.2 kg	Medium, 2.0 kg	Large, 1.2 kg	Large, 2.0 kg
Base: EBITDA a year / month money is earned back	67,227 / m24	124,436 / m17	35,267 / never	871,112 / m54	126,255 / never	2,796,025 / m63
Price −20%	46,243 / m31	89,461 / m20	−274,181 / never	355,365 / m110	−883,188 / never	1,113,620 / never
Price +20%	88,212 / m21	159,411 / m15	344,715 / m112	1,386,858 / m40	1,135,698 / never	4,478,431 / m46
Output −20%	50,065 / m29	95,832 / m20	−215,486 / never	453,189 / m89	−674,676 / never	1,461,140 / m102
Too much supply: price −35% from year 3	30,504 / m24	63,230 / m17	−506,267 / never	−31,444 / never	−1,640,270 / never	−148,183 / never

EBITDA is in CNY a year at full output. "m24" means total cash turns positive in month 24 (within ten years). In the too-much-supply case, prices fall during months 24–36 and then stay 35% lower. The small farm's result does not change. It earns back its money before prices fall.

Other cases

- **土鳖虫 household (500 m²):** start-up cost CNY 66,585. Sales about CNY 95,000 a year. EBITDA about CNY 4,550. It never earns back the money, even with unpaid family work. **土鳖虫 commercial (3,000 m²):** start-up cost CNY 1,076,000. Sales CNY 684,000. EBITDA −CNY 562,500. Six staff cost about CNY 42,000 a month, but monthly sales are only CNY 57,000.
- **American cockroach base (300 t dried a year, CNY 45 million):** cash cost is about CNY 51/kg. Market prices are only CNY 29–50. It needs about CNY 118/kg to earn back the money in five years. Do it only with a contract or joint company with a drug maker.
- **Japan, small farm:** similar equipment costs about JPY 1.5–2.5 million (CNY 63,000–105,000). Shop prices per insect are many times China's wholesale price. But labour (JPY 1,090/hour), power and shipping also cost many times more. So get local prices and run the model before deciding.

Financing

- **Pilot:** our own savings, or CNY 100,000 from one private investor. Another option is a Hebei or Tianjin start-up loan with a government guarantee (创业担保贷款). It gives up to CNY 300,000 per person. Hebei limits the interest rate to LPR + 150 bp (1.5%) and pays half the interest (冀人社规〔2026〕1号).

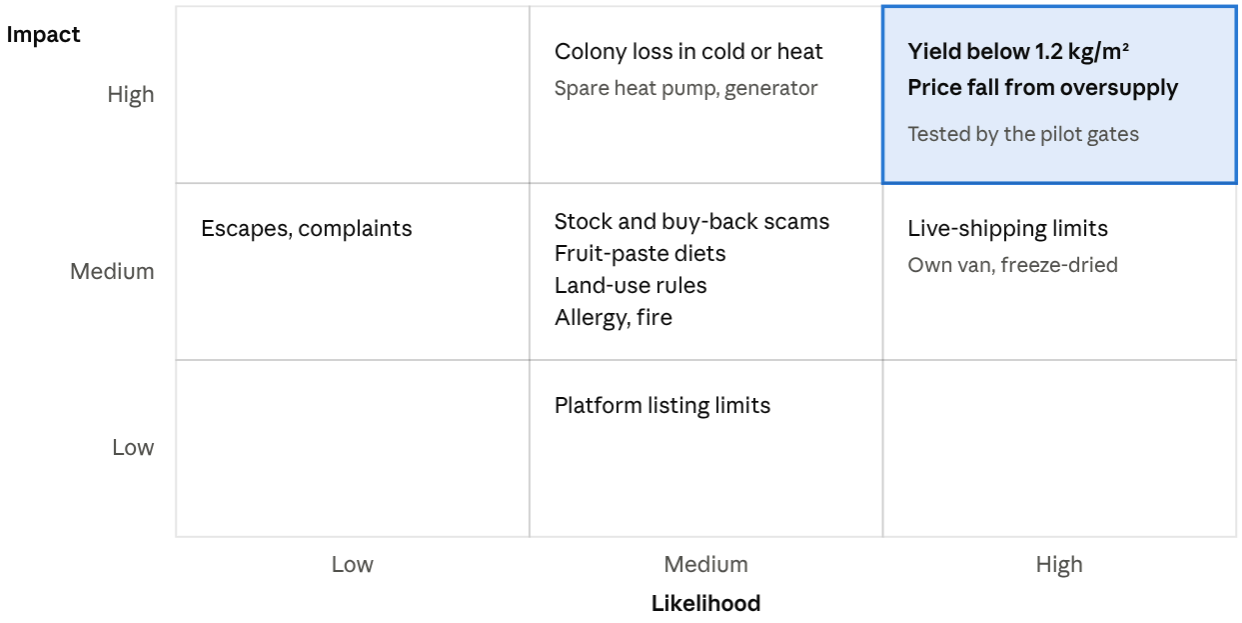
- **Medium farm:** at least half of the CNY 2.5 million comes from investors as shares. The rest comes from guaranteed loans for small firms and from late payment to suppliers. In Hebei and Tianjin these loans give up to CNY 4 million for up to two years. Before building starts, buyers must sign for most of the output.
- **Large farm:** only with a long-term partner, such as a buyer of feed or freeze-dried pet food. This partner must sign a contract to buy most of the output.

Risks

Two risks can make an investor lose the money. Both are business risks. One is output below 1.2 kg/m². The other is a price fall because there is too much supply. Climate and shipping risks are real, but good equipment and planning can reduce them. We avoid sanctions and land rules by choosing the right site.

Yield and price are the two high-likelihood, high-impact risks

This plan's judgement of each risk for a northern Hebei dubia farm



Not placed: North Korea sanctions, avoided by not operating there.

Risk matrix · 10 risks by likelihood and impact (this plan's judgement)

Risk	Likelihood	Impact	Evidence	What we do about it
Output lower than planned	High	High	The only farm figure we found means about 0.9 kg/m² a month. Medium-farm EBITDA turns negative below about 1.17 kg/m²	Prove output in the pilot. Build the medium farm only if output reaches 1.9 kg/m²
Prices fall sharply because of too much supply	High	High	土鳖虫 fell from CNY 125 to 35–45/kg in four years. Dubia buy-back schemes are spreading. Some dubia sizes fell from over CNY 100 to tens of yuan per jin	Sell to many customers. Keep costs low. Test the model with prices 35% lower
Insects die in winter cold or summer heat	Medium	High	Young dubia die at 35 °C. Winters go below −20 °C. Beijing reached 41 °C in June 2023	Heat pumps with spare units, alarms, generator, separate rooms
Limits on shipping live animals	High	Medium	Postal rules limit live animals. Reports say SF stopped carrying them. 13% of parcels take over 72 hours	Deliver with our own van. Sell freeze-dried products
Scams that sell breeding insects with a promise to buy back	Medium	Medium	The government named these scams in 2026	Pay market prices. Never accept buy-back deals
Owners switch to fruit-paste food	Medium	Medium	Ready-made powder food (MRP) is growing for crested and gargoyle geckos	Sell to owners of insect-eating animals: bearded dragons, leopard geckos

Risk	Likelihood	Impact	Evidence	What we do about it
Insects escape and neighbours complain	Low	Medium	Escape at Dafeng in 2013. Village complaints in Jinan	Sealed rooms. A site away from homes
Rules on land use	Medium	Medium	The 2026 rules on farm buildings on farmland depend on each province's list. No province lists insects yet	Rent an existing building. Get written approval before signing the lease
Online shops limit listings	Medium	Low	1688 searches for "蟑螂" found nothing	Sell as 杜比亚. Build direct customer accounts
Sanctions (North Korea)	Certain if we try	High	UN bans on joint companies, sending machines and exports	Do not operate there
Allergy, fire	Medium	Medium	Allergy-causing dust in the air. Heaters near cardboard egg trays	Fresh air, face masks, checked wiring, smoke detectors

Likelihood and impact are our own judgement.

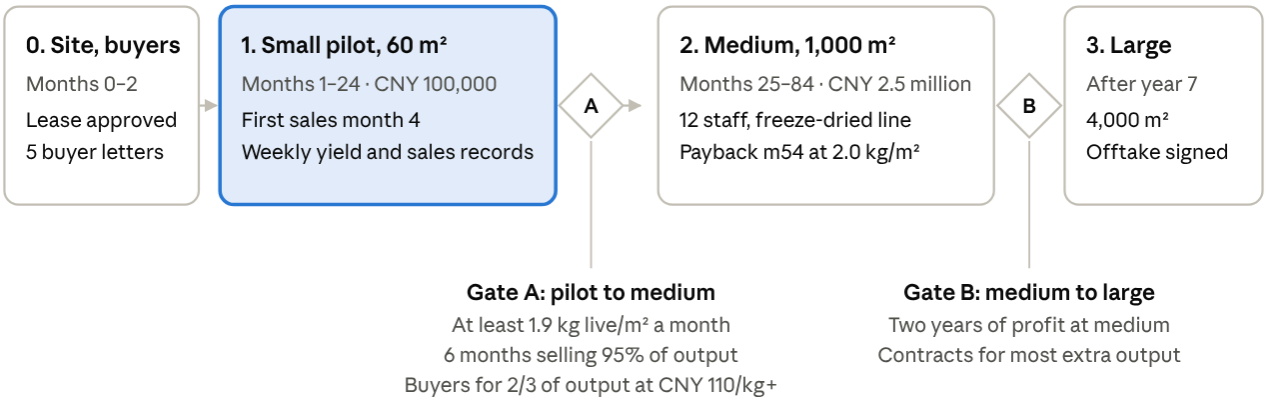
Permits to check: a business licence that includes insect farming. The land use of the site, and any province filing for farm buildings on farmland. Whether we must file an environmental report. The farm power price, from the local 供电所. A claim for VAT exemption. A feed licence before we sell freeze-dried products. Fertiliser registration for insect droppings (frass). Written agreement from the landlord. Insurance. In Japan: business registration and the rules of the prefecture.

Strategy

We build in three steps. Each step starts only when the step before has passed a written test. First, a pilot proves output and buyers. Second, a medium farm proves profit at a larger size. Third, we build a large farm only with contracts. Output is the key unknown. Until we measure it, only about CNY 100,000 is at risk.

Each phase is released only after the previous gate is passed

Phases not to scale; months from the start of the project



Stop rule: below 1.2 kg/m² by month 18, keep the pilot as a sideline and raise no phase 2 money.

Roadmap · 4 phases, 2 gates

Timeline in steps

Step	Months	Main work	Money at risk	Test to move on
0. Site and buyers	0–2	Choose a site in northern Hebei or outer Tianjin. It needs three-phase power, no close neighbours and written land-use approval. Find five or more Beijing–Tianjin pet shops and breeders as first buyers	under CNY 5,000	Site lease signed and letters from five buyers
1. Small pilot	1–24	Prepare 60 m². Buy insects in stages. First sales in month 4. Full output in month 15. Every week, record young per female, survival, feed, power and sales	CNY 100,000	Test A below

Step	Months	Main work	Money at risk	Test to move on
2. Medium farm	25–84	Rent 1,300 m². Prepare 1,000 m². Hire 12 people. First sales in month 5 of this step. Full output in month 26. Add freeze-dried products when the licence question is answered	CNY 2.5 million	Test B below
3. Large farm	after year 7	Build 4,000 m² only with signed contracts to buy the output	about CNY 9 million at the highest point	—

Tests that must be passed

- **Test A, pilot to medium:** six months of selling 95% of output. At least 7.5 young insects for sale per female a month. Output of at least 1.9 kg live per m² a month. (In the model, the medium farm needs this to earn back its money in five years.) Buyers for two-thirds of medium output at an average of CNY 110/kg or more.
- **Test B, medium to large:** two years of profit at medium size, and contracts for most of the extra output.
- **Stop rule:** if the pilot cannot reach 1.2 kg/m² by month 18, the owner keeps it as a small side business. We do not raise money for step 2.

How we will sell

- **Sell on reliability:** the same sizes every time, supply all year, a promise that insects arrive alive, and delivery in our own van. In winter, other farms raise prices because heating costs more. A well-heated farm can win customers then.
- **Channels:** Taobao, Pinduoduo, Douyin and Xianyu shops. WeChat groups. The 平谷 reptile fair (June) and Tianjin reptile shows. Beijing pet markets. Gecko breeders and zoos (a Wuhan zoo buys dubia every month).
- **Year 2–3 options:** freeze-dried dubia for buyers far away. A small 土鳖虫 side business, but only with spare rooms in the countryside and family workers.

Sources

AI research agents opened the source pages on 8 November 2026. Links are given beside each table. The main sources are:

- **Markets and prices:** 8u58.com 土鳖虫 and 蟑螂 market reports; 药通网 via Sohu (土鳖虫 price history, 2025-08-25); 康美中药网 and 一亩田 (search snippets); 1688 and 爱企查 seller pages (dubia); 2025 小宠异宠白皮书 via 宠业家; 米内网 (Kangfuxin sales).
- **Farms:** The Paper, 红星新闻 and Kelun (American cockroach bases); 新华网山东 and 大众网 (Caoxian); 腾讯新闻, 现代快报 and 韩城新闻网 (Suining dubia farm); 九原区融媒体中心 via Sina (Baotou).
- **Rules:** MARA (feed catalogue No. 1028, invasive-species measures, Livestock Law); Biosecurity Law; VAT Law implementing rules; 自然资发〔2026〕146号 (facility farmland); 邮政法实施细则; Japan's Ministry of the Environment and MAFF; UN Security Council resolutions 2375 and 2397 via Wikipedia.
- **Costs:** Mysteel (feed prices); Beijing, Tianjin and Hebei price bureaus (power tariffs); MOHRSS and Hebei 人社厅 (minimum wages, start-up loans); 58同城 and 土流网 (rent and pay ads); 1688 (equipment).
- **Climate:** CMA normals via Wikipedia climate boxes; KMA normals via Wikipedia; JMA normals.
- **Science:** Oonincx et al. 2015 (feed conversion); Baidu Baike and hobby guides (biology).

Many Chinese feeder-insect prices came from search result snippets, because shop sites were blocked. The financial figures are estimates from this plan's model. An AI stress test ran them again at northern Hebei costs. An AI fact-check compared the claims with the research reports. We used their corrections. No human expert has checked them. Ask a local farmer, accountant and lawyer to check the plan before you invest. Photos are from Wikimedia Commons, Flickr and iNaturalist, under the licences named in each caption.